

An efficient 3-D curvilinear structured grid generation method for large-scale seismic wave simulation in regions with irregular topography

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SUMMARY

The finite difference method (FDM) based on curvilinear structured grids is an efficient numerical algorithm that retains the advantages of conventional FDM, such as computational efficiency and ease of implementation, while effectively simulating seismic wave propagation in regions with irregular topography. Grid quality plays a crucial role in the finite difference simulations, as it directly affects the reliability, stability and efficiency of the simulation. Moreover, large-scale 3-D simulations often require billion-point grids, making the efficient generation of high-quality grids a significant challenge. The curvilinear grid used in seismic wave simulations has a special feature compared to other computational problems: only the shape of the free surface is fixed, while the other boundaries remain flexible. Based on this feature, we developed a straightforward parabolic grid generation method that starts from the free surface and expands the grid layer-by-layer along the depth direction towards the pre-defined bottom boundary. Additionally, this method is highly efficient, easy to parallelize and capable of rapidly generating large-scale grids. We tested the grid generation method and its application in seismic wave simulations using five irregular topography models. Grid quality assessments demonstrate that the proposed method efficiently generates high-quality grids. Furthermore, seismic wave simulations reveal that, compared to vertically stretched grids, the generated grids reduce computation time while maintaining sufficient simulation accuracy.

Key words: Numerical modelling, Computational seismology, Wave propagation.

1. INTRODUCTION

The numerical simulation of seismic wave propagation uses computer models to simulate the propagation of seismic waves in sub-surface media, serving as a vital tool in seismological research. Topography can significantly impact seismic wave propagation by causing wave scattering and amplifying wave amplitudes. Therefore, achieving accurate simulations of seismic wave propagation in regions with irregular topography is essential.

As there is no analytical solution for 3-D seismic wave propagation in a complex medium, numerical simulations remain the primary method for studying and understanding seismic wave behaviour. Various numerical methods, including those based on unstructured grids like the finite element method (FEM; Marfurt 1984; Bao *et al.* 1998), spectral element method (SEM; Priolo *et al.* 1994; Komatitsch & Vilotte 1998; Komatitsch & Tromp 1999), and discontinuous Galerkin method (DGM; Dumbser & Kaser 2006; Kaser & Dumbser 2006; He *et al.* 2019; Cao *et al.* 2024), are well-suited for seismic wave simulation in regions with irregular topography due

to their flexibility in describing complex geometries. However, generating unstructured grids is labour-intensive and time-consuming (Almuhaidib & Toksöz 2015).

In contrast, FDM (Boore 1972; Kelly *et al.* 1976; Virieux 1986; Levander 1988; Zahradnik 1995; Graves 1996; Moczo *et al.* 2002), which relies on structured grids, remains popular for seismic wave simulation due to its simplicity, efficiency and ease of parallel computing. However, the traditional FDM is limited to Cartesian coordinates, making it less suitable for modelling irregular topography. To address this limitation, various forms of FDM, including those based on vertically stretched grids (Hestholm & Ruud 1994, 1998; Tarras *et al.* 2011; de la Puente *et al.* 2014; Shragge & Konuk 2020) and curvilinear grids (Zhang & Chen 2006; Ely *et al.* 2008; Appelo & Petersson 2009; Lan & Zhang 2011; Zhang *et al.* 2012a; Shragge 2014b; Solano *et al.* 2016; Rao & Wang 2018; O'Reilly *et al.* 2022; Zheng *et al.* 2023), have been developed to better handle geometrically complex topography. The quality of the grid, particularly the grid step size, orthogonality and smoothness, has a significant impact on the accuracy and efficiency of seismic wave

numerical simulations (Ely *et al.* 2009; Rao & Wang 2013; Zhang *et al.* 2014). Additionally, the expansion of the simulation area and the increase in simulation frequency result in a substantial increase in the number of grids. Therefore, an efficient and straightforward generation of large-scale, high-quality structured grids is essential for applying FDM to seismic wave simulation in regions with irregular topography.

Vertically stretched grids, a simplified form of general curvilinear grids, are widely used for seismic wave simulations in regions with irregular topography. Although these grids can be generated quickly, they exhibit poor orthogonality near the undulating free surface. Additionally, vertically stretched grids cannot represent all topographies, such as vertical dams or overhangs.

The most popular methods for generating general curvilinear grids include algebraic and partial differential equation (PDE) grid generation methods. Algebraic methods are very fast but are only suitable for simple geometries or as an initial grid for elliptic grid generation method. PDE-based methods can be categorized into three types based on the governing equations: elliptic, parabolic and hyperbolic. The elliptic method requires precise specification of all boundary point positions and iteratively optimizes positions of the grid points until the misfit error is achieved. In contrast, the hyperbolic method starts from a given free surface and expands the grid layer by layer in the depth direction. Similarly, the parabolic method begins at the free surface and extends the grid downward, layer by layer, towards a pre-defined bottom boundary. Since both the hyperbolic and parabolic methods generate grids layer by layer, they offer significantly higher grid generation efficiency, often requiring one to two orders of magnitude less computational time compared to elliptic methods (Chan & Steger 1992).

Several studies have developed 2-D general curvilinear grid generation methods for seismic wave simulations. Sun & Jiang (2009) used an elliptic equation to generate orthogonal body-fitted curvilinear grids, with the source terms controlled by the Hilgenstock method. Rao & Wang (2013) employed an elliptic equation for grid generation, improving grid orthogonality by adjusting the boundary points. Wang *et al.* (2015) applied an elliptic equation to generate body-fitted curvilinear grids for reverse-time wavefield migration. Li *et al.* (2024) used the hyperbolic grid generation method to generate high-quality grids for extreme 2-D topography. Shragge (2014a) utilized conformal mapping to generate 2-D orthogonal body-fitted grids.

Elliptic methods are often unsuitable for the rapid generation of large-scale 3-D grids due to their time-consuming iterative processes. Similarly, the conformal mapping method is restricted to 2-D applications, limiting its use in more complex 3-D geological structures. Seismic wave simulation grids have a special feature compared to other computational problems: only the shape of the free surface is fixed, while the other boundary shapes are free to be set. This feature aligns with the characteristics of the parabolic grid generation method. Originally developed in the field of computational fluid dynamics (Nakamura 1982a,b; Edwards 1985; Noack 1985; Zhang *et al.* 2012b), the parabolic grid generation method progressively generates the grid layer by layer, starting from a given inner boundary and extending outwards to the specified outer boundary. Furthermore, the parabolic grid generation method can effectively implement parallel computing, significantly expanding the scale and making it particularly suitable for large-scale grids.

Algebraic grid prediction is a key step in the parabolic grid generation method. However, the clustering factor and switching factor calculation formulae in the algebraic prediction method proposed

by Noack (1985) have certain limitations, making it difficult to generate structured grids in some cases (Zhang *et al.* 2012b). Zhang *et al.* (2012b) improved this method, but their enhanced factor calculation approach relies on information from the side boundary grid points, requiring the positions of all boundary grid points to be pre-defined. We have improved the calculation method for these factors, using the input normalized grid step size in the depth direction to calculate these factors. This allows for flexible control over the grid generation step size, eliminating the need to define side boundary grid points.

In this paper, we implement an efficient and straightforward parabolic method and develop a parallel computing program to generate large-scale, high-quality grids suitable for FDM seismic wave simulations in regions with irregular topography. We tested the effectiveness of the grid generation method using five irregular topography models. We validated the applicability of the generated grids in seismic wave simulations by employing the curvilinear grid finite difference method (CGFDM; Zhang *et al.* 2012a) as the seismic wave simulation solver

2. PARABOLIC GENERATION METHOD FOR LARGE-SCALE 3-D CURVILINEAR GRID

2.1. Grid governing equation

A point of the 3-D curvilinear grid has two types of coordinates: the Cartesian coordinate (x, y, z) , and the computational coordinate or the generalized coordinate $\xi(x, y, z)$, $\eta(x, y, z)$ and $\zeta(x, y, z)$. The mapping between the two coordinates (Fig. 1) can be expressed as

$$\begin{aligned} x &= x(\xi, \eta, \zeta), \\ y &= y(\xi, \eta, \zeta), \\ z &= z(\xi, \eta, \zeta). \end{aligned} \quad (1)$$

The grid governing equations are Laplace equations in the computational domain (Thomas 1982)

$$\alpha_1 f_{,\xi\xi} + \alpha_2 f_{,\eta\eta} + \alpha_3 f_{,\zeta\zeta} + 2(\beta_{12} f_{,\xi\eta} + \beta_{23} f_{,\eta\zeta} + \beta_{13} f_{,\xi\zeta}) = 0, \quad (2)$$

where $f = x, y$ or z ,

$$\alpha_1 = |\vec{r}_{,\eta}|^2 |\vec{r}_{,\zeta}|^2 - (\vec{r}_{,\eta} \cdot \vec{r}_{,\zeta})^2, \quad (3)$$

$$\alpha_2 = |\vec{r}_{,\xi}|^2 |\vec{r}_{,\zeta}|^2 - (\vec{r}_{,\xi} \cdot \vec{r}_{,\zeta})^2, \quad (4)$$

$$\alpha_3 = |\vec{r}_{,\xi}|^2 |\vec{r}_{,\eta}|^2 - (\vec{r}_{,\xi} \cdot \vec{r}_{,\eta})^2, \quad (5)$$

$$\beta_{12} = (\vec{r}_{,\xi} \cdot \vec{r}_{,\zeta})(\vec{r}_{,\eta} \cdot \vec{r}_{,\zeta}) - (\vec{r}_{,\xi} \cdot \vec{r}_{,\eta}) |\vec{r}_{,\zeta}|^2, \quad (6)$$

$$\beta_{23} = (\vec{r}_{,\xi} \cdot \vec{r}_{,\eta})(\vec{r}_{,\xi} \cdot \vec{r}_{,\zeta}) - (\vec{r}_{,\eta} \cdot \vec{r}_{,\zeta}) |\vec{r}_{,\xi}|^2, \quad (7)$$

$$\beta_{13} = (\vec{r}_{,\eta} \cdot \vec{r}_{,\zeta})(\vec{r}_{,\xi} \cdot \vec{r}_{,\eta}) - (\vec{r}_{,\xi} \cdot \vec{r}_{,\zeta}) |\vec{r}_{,\eta}|^2, \quad (8)$$

$$\vec{r} = (x, y, z)^T, \quad (9)$$

a comma in subscripts means a derivative with respect to the variable and the superscript T represents the transpose operator.

Let I , J and K be the total number of grid points in ξ , η and ζ directions, respectively. Grid generation starts from the free surface, with negative direction of the axis ζ as the direction of advancement. The index $k = 1$ represents the free surface, while the index $k = K$ represents the bottom surface. The grid distributions of these two

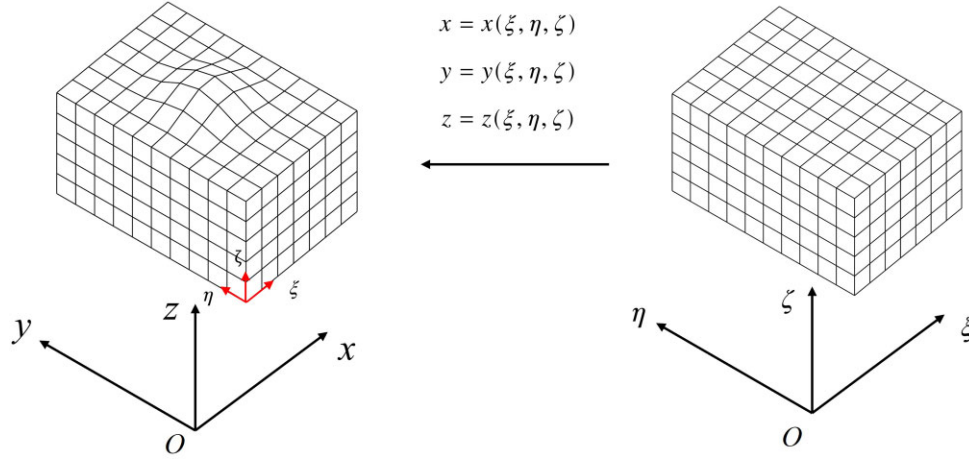


Figure 1. Mapping (left) between the boundary-conforming grid in the physical space (x, y, z) and (right) the uniform grid in the computational space (ξ, η, ζ).

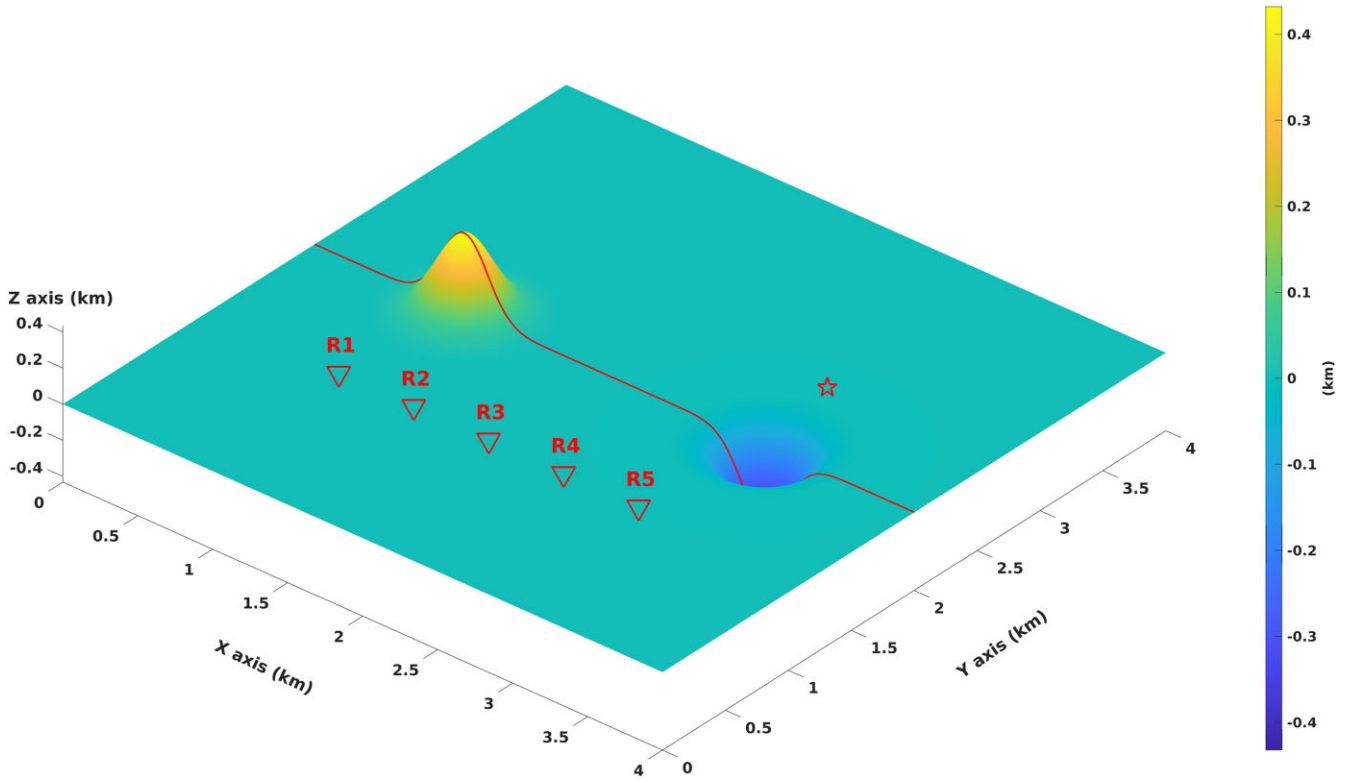


Figure 2. Illustration of the Gaussian peak-valley model topography, where the asterisk represents the seismic source, the inverted triangle represents the receivers on the free surface and the curve represents the surface position of the grid slice to be displayed.

surfaces are user-specified. We assume that the surface with index k represents the surface being generated, all the surface with indices less than k are already generated, while those with indices greater than k are yet to be generated.

2.2. Algebraic grid prediction

To generate k surface grids, first we use an algebraic prediction method to predict temporary locations of the grid points at k and $k + 1$ surfaces, which are required to compute the derivatives $f_{,\xi\xi}$, $f_{,\eta\eta}$, $f_{,\xi\xi}$, $f_{,\xi\eta}$, $f_{,\xi\zeta}$, $f_{,\eta\zeta}$ and the coefficients α_1 , α_2 , α_3 , β_{12} , β_{13} , β_{23} .

Let $R_{i,j,k-1}$ be the distance from point $\vec{r}_{i,j,k-1}$ to the point $\vec{r}_{i,j,K}$. $\vec{R}_{i,j,k-1}$ is the corresponding vector.

$$\vec{R}_{i,j,k-1} = \vec{r}_{i,j,K} - \vec{r}_{i,j,k-1}. \quad (10)$$

The unit normal vector $\vec{n}$ of point $\vec{r}_{i,j,k-1}$ is

$$\vec{n} = -\frac{\vec{r}_{\xi} \times \vec{r}_{\eta}}{|\vec{r}_{\xi} \times \vec{r}_{\eta}|}. \quad (11)$$

The predicted point $(\vec{r}_o)_{i,j,k+1}$ along this normal vector are given by

$$(\vec{r}_o)_{i,j,k+1} = \vec{r}_{i,j,k-1} + \vec{n}\varepsilon_c R_{i,j,k-1}, \quad (12)$$

where ε_c is a clustering factor.

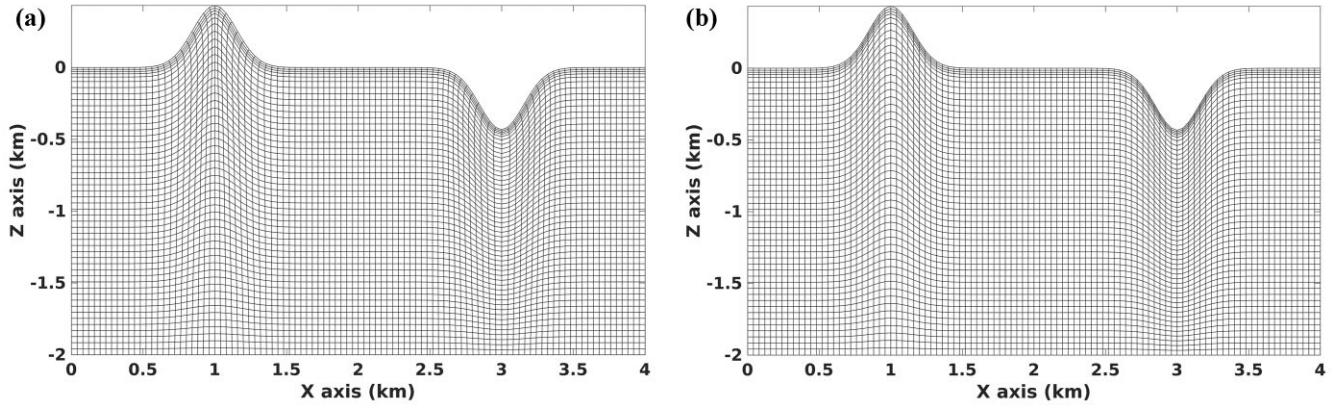


Figure 3. The Gaussian peak-valley model grid slice generation illustration. For clarity, the grid density has been reduced. (a) The curvilinear grid. (b) The vertically stretched grid.

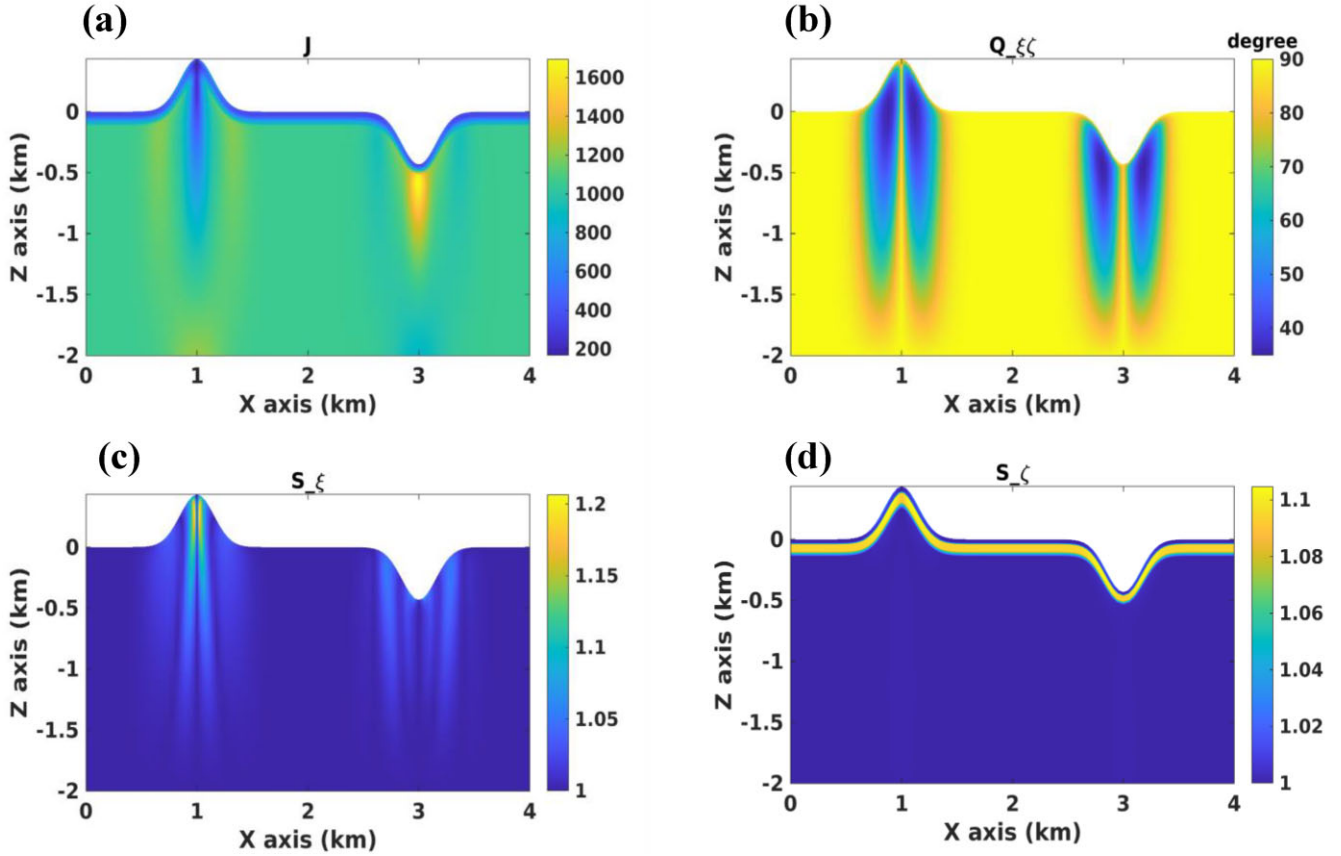


Figure 4. The Gaussian peak-valley model grid slice quality. (a) Jacobian determinant J . (b) Grid orthogonality $Q_{\xi\zeta}$. (c) Grid smoothness S_{ξ} . (d) Grid smoothness S_{ζ} .

If the predicted grid is advanced only along the orthogonal direction from the irregular free surface, the grid line may converge and cross. Therefore, another point located on a straight line between the point $\vec{r}_{i,j,k-1}$ and the point $\vec{r}_{i,j,K}$ is needed to modify the point on the orthogonal line. Another predicted point on the straight line is obtained by the following linear interpolation,

$$(\vec{r}_s)_{i,j,k+1} = \vec{r}_{i,j,k-1} + \varepsilon_c \vec{R}_{i,j,k-1}. \quad (13)$$

The final predicted point is a blend of the above two points,

$$\vec{r}_{i,j,k+1} = \varepsilon_s (\vec{r}_o)_{i,j,k+1} + (1 - \varepsilon_s) (\vec{r}_s)_{i,j,k+1}, \quad (14)$$

where ε_s is a switching factor, the formula is

$$\begin{aligned} \zeta &= (k-1)/(K-1), \\ \varepsilon_s &= e^{-a\zeta}, \end{aligned} \quad (15)$$

where a is a damping coefficient, and its value is greater than 0.

When the value of k is smaller, ε_s approaches 1 and the predicted points $\vec{r}_{i,j,k+1}$ are primarily determined by orthogonal direction points $(\vec{r}_o)_{i,j,k+1}$. As k increases, ε_s decays exponentially and the predicted points $\vec{r}_{i,j,k+1}$ are predominantly determined by straight line direction points $(\vec{r}_s)_{i,j,k+1}$. The damping coefficient a controls the orthogonality of the grid near the free surface in the generated

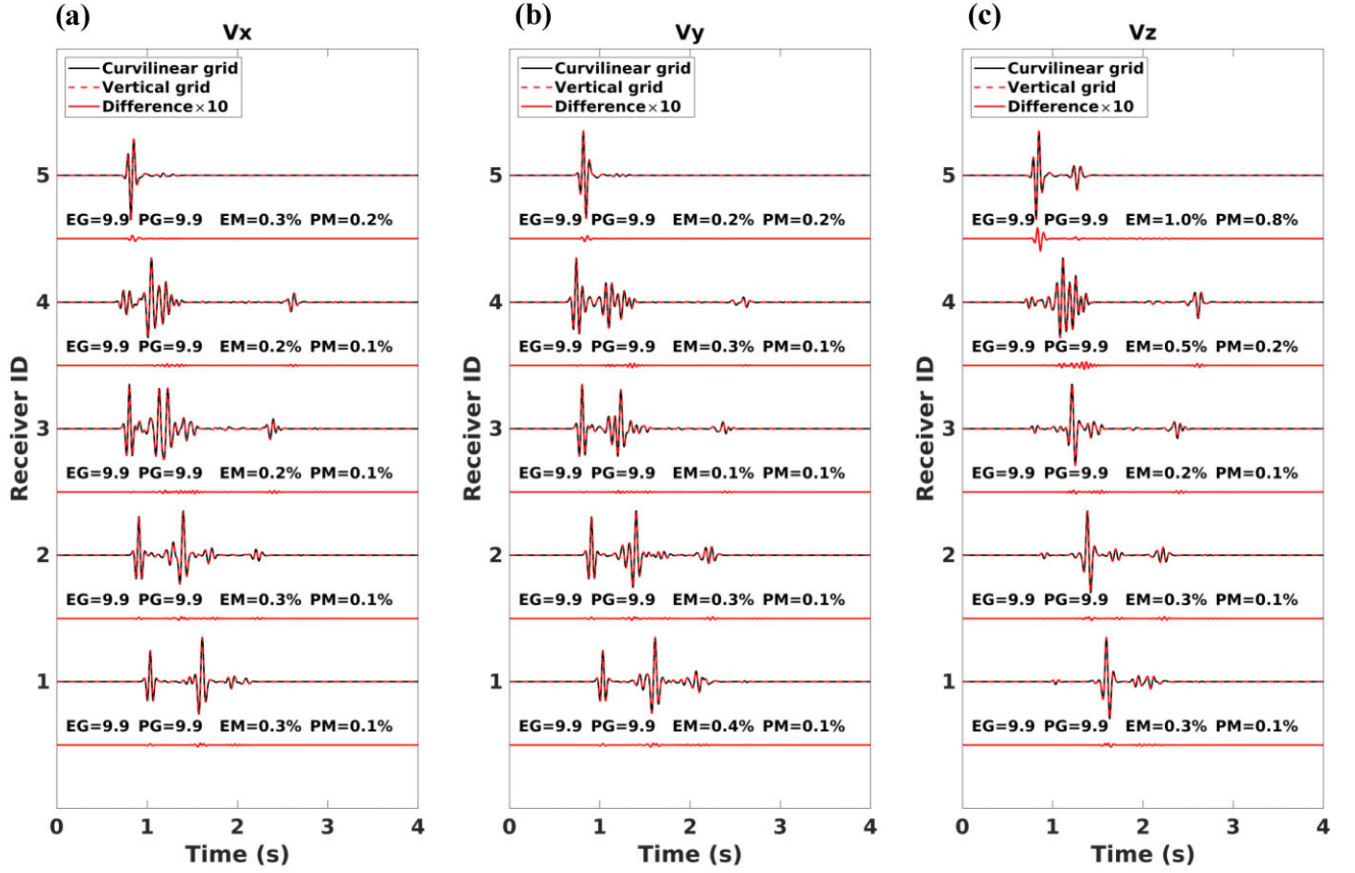


Figure 5. Comparison of the particle velocity components of the Gaussian peak-valley model at different receivers. (a) The x-component. (b) The y-component. (c) The z-component.

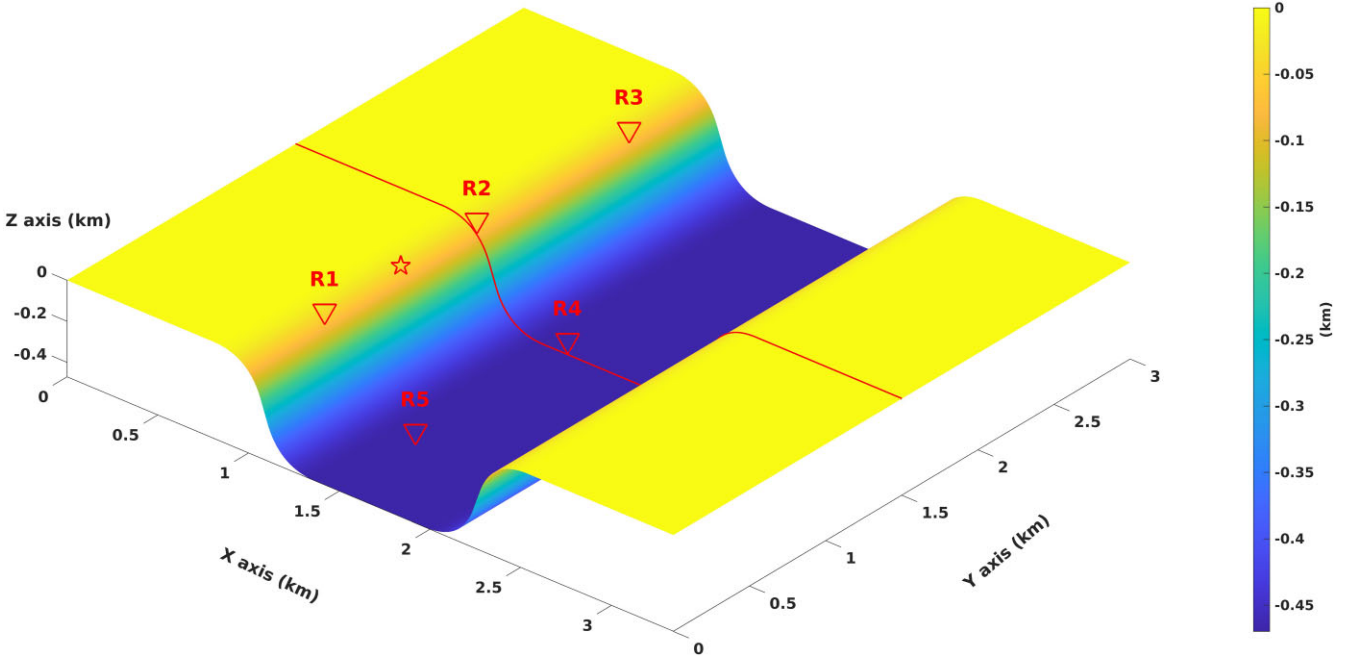


Figure 6. Illustration of the concave model topography, where the asterisk represents the seismic source, the inverted triangle represents the receivers on the free surface and the curve represents the surface position of the grid slice to be displayed.

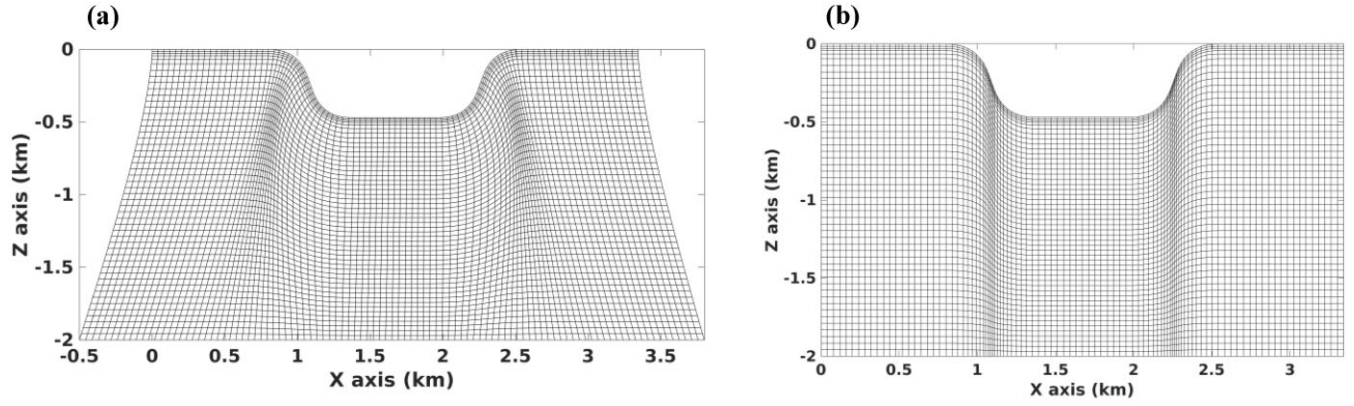


Figure 7. The concave model grid slice generation illustration. For clarity, the grid density has been reduced. (a) The curvilinear grid. (b) The vertically stretched grid.

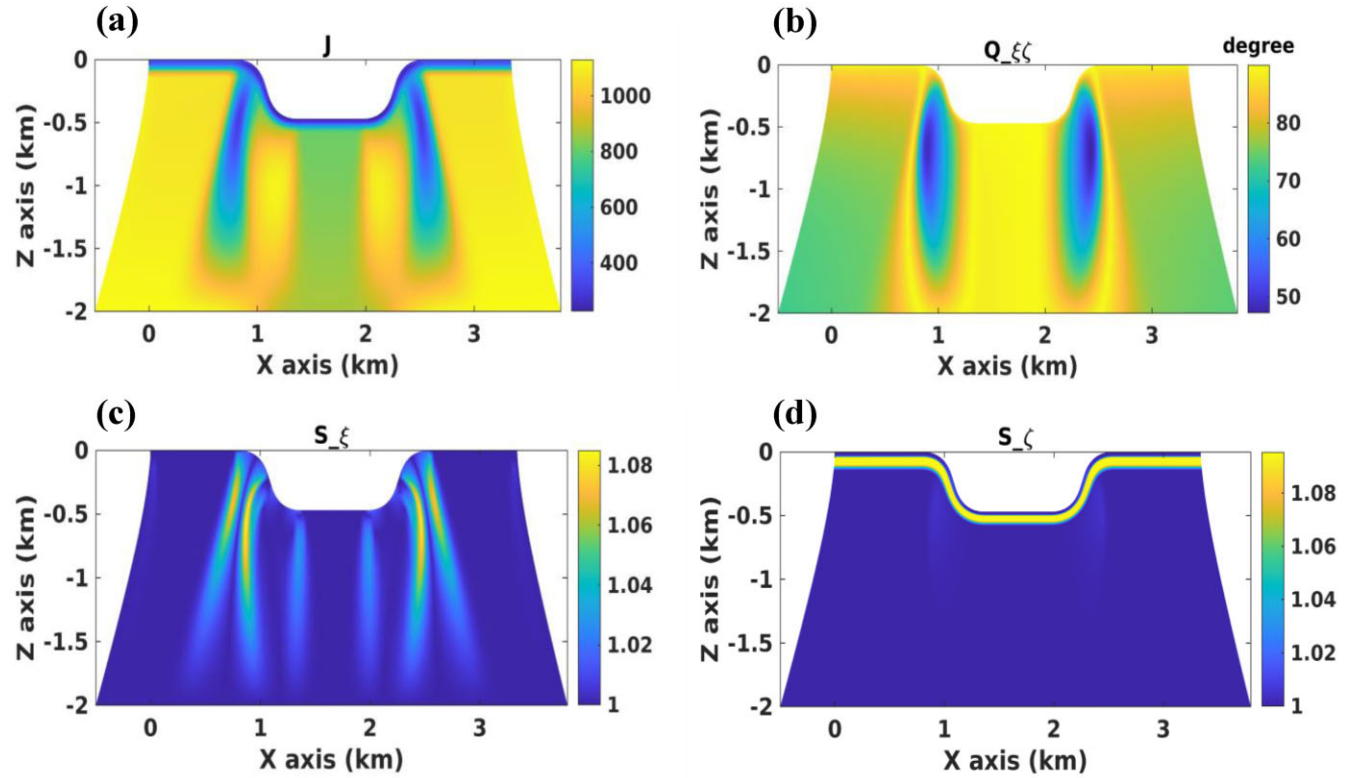


Figure 8. The concave model grid slice quality. (a) Jacobian determinant J . (b) Grid orthogonality $Q_{\xi\xi}$. (c) Grid smoothness S_{ξ} . (d) Grid smoothness S_{ζ} .

grid. The smaller the value of a , the slower the exponential decay of ε_s , resulting in better grid orthogonality near the free surface, but it could reduce the smoothness of the grid and even cause grid intersection. The value of a is generally between 5 and 50.

The k surface points are predicted based on the predicted $k+1$ surface points,

$$\vec{r}_{i,j,k} = \vec{r}_{i,j,k-1} + \varepsilon_i(\vec{r}_{i,j,k+1} - \vec{r}_{i,j,k-1}), \quad (16)$$

where ε_i is an interpolation factor.

From eq. (13), it can be observed that, ε_c represents the ratio of the distance from the $k-1$ layer to the $k+1$ layer to the distance from the $k-1$ layer to the K layer. The normalized length d_k of different grid layers can be calculated using the input normalized

grid step size in the depth direction.

$$\begin{aligned} d_1 &= 0, \\ d_k &= d_{k-1} + h_{k-1}, \quad k = 2, 3, \dots, K, \end{aligned} \quad (17)$$

where h_{k-1} is the input normalized grid step size of the $k-1$ surface.

Therefore, the ε_c formula is

$$\varepsilon_c = (d_{k+1} - d_{k-1}) / (d_K - d_{k-1}). \quad (18)$$

Similarly, the ε_i formula is

$$\varepsilon_i = (d_k - d_{k-1}) / (d_{k+1} - d_{k-1}). \quad (19)$$

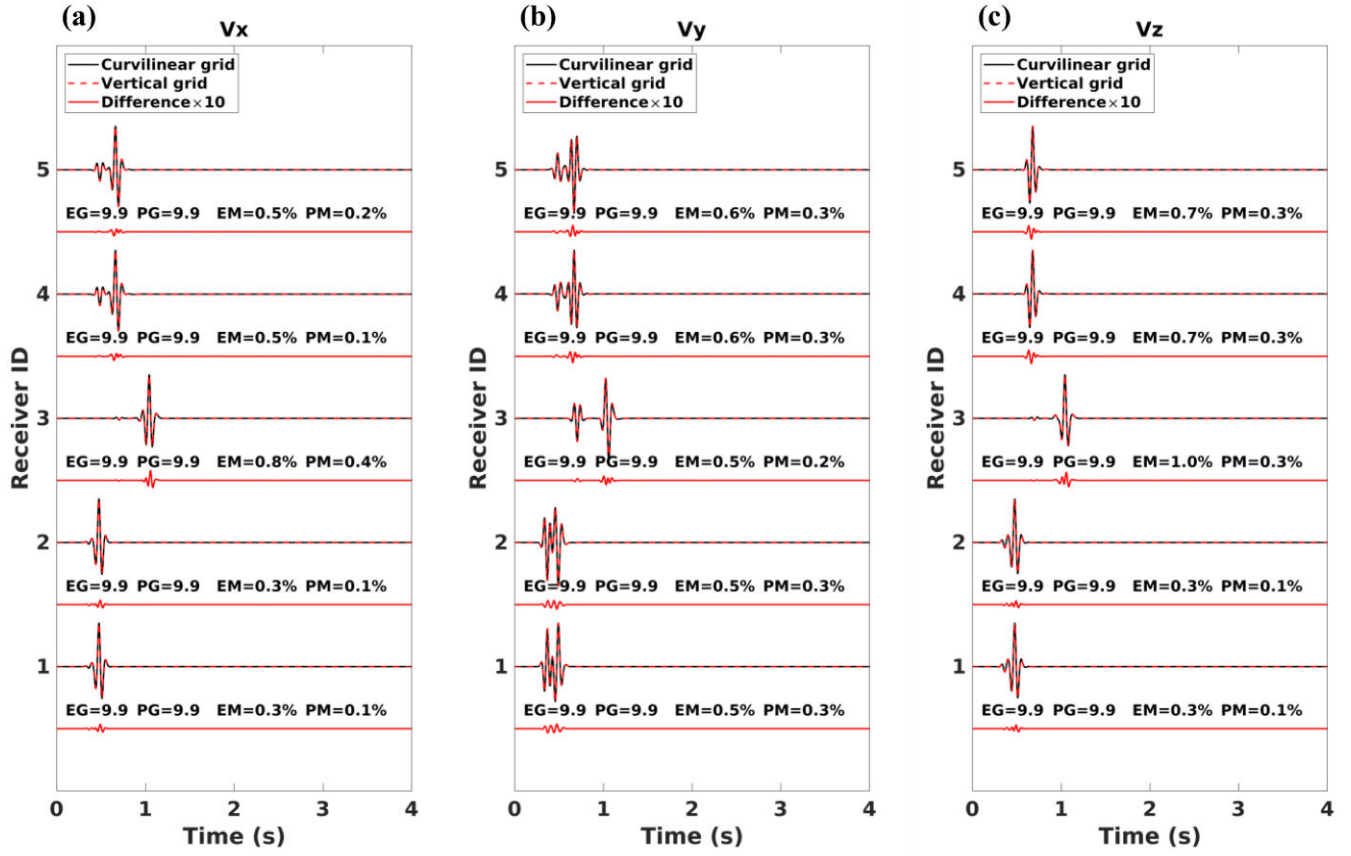


Figure 9. Comparison of the particle velocity components of the concave model at different receivers. (a) The x-component. (b) The y-component. (c) The z-component.

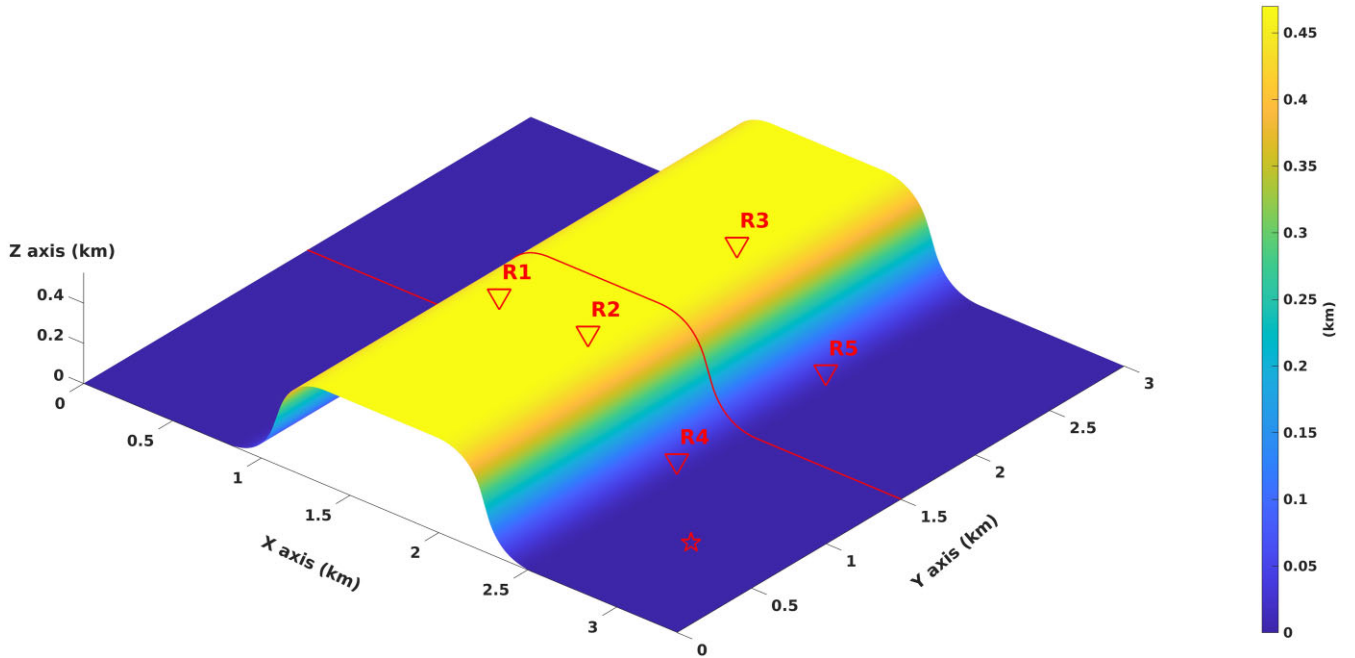


Figure 10. Illustration of the convex model topography, where the asterisk represents the seismic source, the inverted triangle represents the receivers on the free surface and the curve represents the surface position of the grid slice to be displayed.

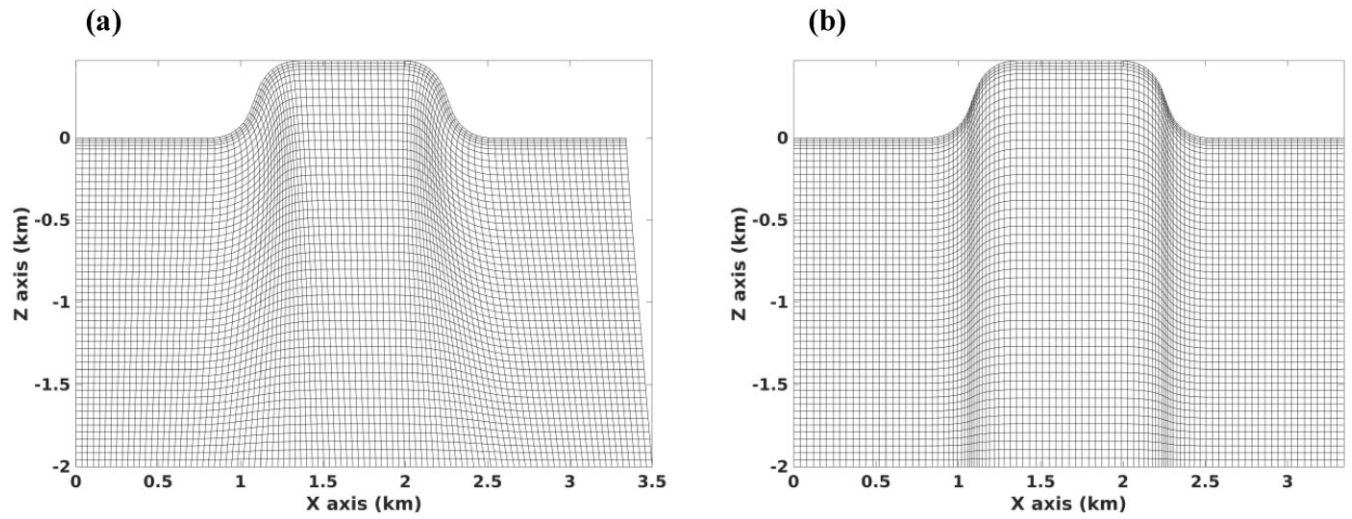


Figure 11. The convex model grid slice generation illustration. For clarity, the grid density has been reduced. (a) The curvilinear grid. (b) The vertically stretched grid.

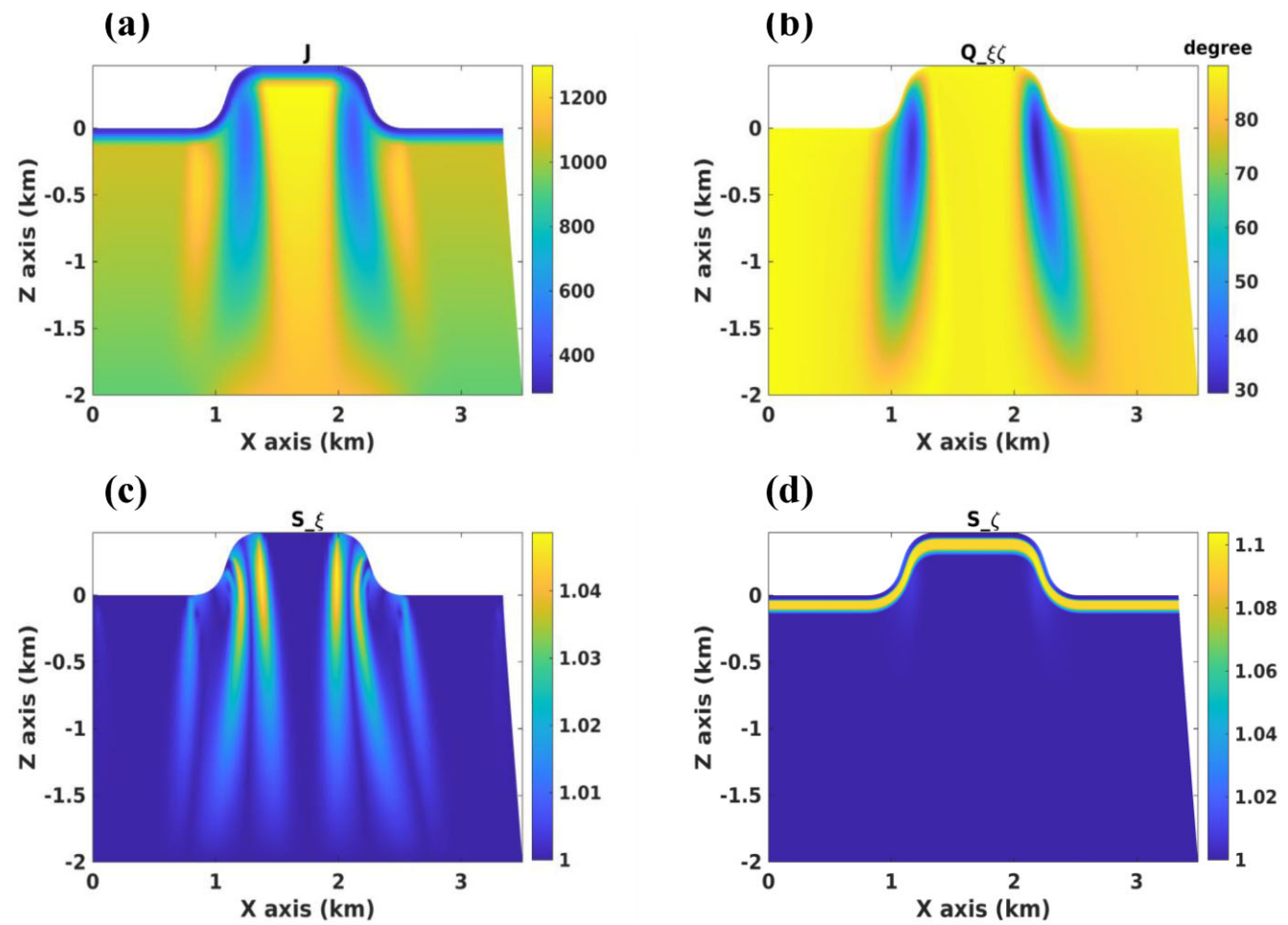


Figure 12. The convex model grid slice quality. (a) Jacobian determinant J . (b) Grid orthogonality $Q_{\xi\zeta}$. (c) Grid smoothness S_{ξ} . (d) Grid smoothness S_{ζ} .

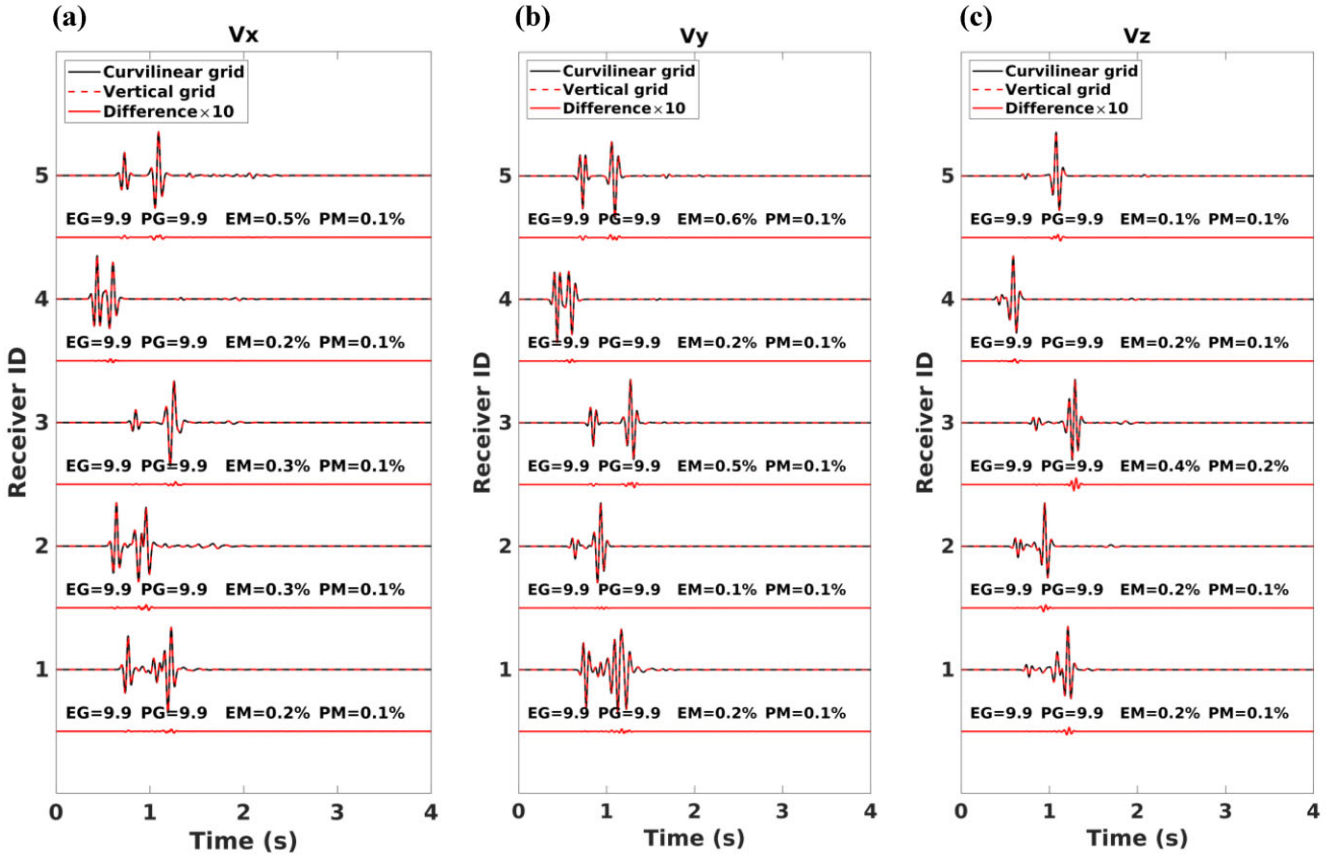


Figure 13. Comparison of the particle velocity components of the convex model at different receivers. (a) The x-component. (b) The y-component. (c) The z-component.

2.3. Discretized governing equation

Assuming $\Delta\xi = \Delta\eta = \Delta\zeta = 1$, without loss of generality, that all derivatives are approximated by three-point central difference scheme, substituting into eq. (2) at point (i, j, k) gives

$$\begin{aligned} & \alpha_1(f_{i+1,j,k} - 2f_{i,j,k} + f_{i-1,j,k}) \\ & + \alpha_2(f_{i,j+1,k} - 2f_{i,j,k} + f_{i,j-1,k}) + \alpha_3(f_{i,j,k+1} - 2f_{i,j,k} + f_{i,j,k-1}) \\ & + 2\beta_{12}(f_{i+1,j+1,k} + f_{i-1,j-1,k} - f_{i-1,j+1,k} - f_{i+1,j-1,k})/4 \\ & + 2\beta_{23}(f_{i,j+1,k+1} + f_{i,j-1,k-1} - f_{i,j-1,k+1} - f_{i,j+1,k-1})/4 \\ & + 2\beta_{13}(f_{i+1,j,k+1} + f_{i-1,j,k-1} - f_{i-1,j,k+1} - f_{i+1,j,k-1})/4 = 0. \end{aligned} \quad (20)$$

We first solve for the coordinates along ξ direction by moving $f_{i-1,j,k}$, $f_{i,j,k}$ and $f_{i+1,j,k}$ on the left-hand side as unknown variable and all the other variables to the right-hand side as known variable, resulting in the following equation,

$$\begin{aligned} & \alpha_1 f_{i-1,j,k} - 2(\alpha_1 + \alpha_2 + \alpha_3)f_{i,j,k} + \alpha_1 f_{i+1,j,k} = \\ & -\alpha_2(f_{i,j+1,k} + f_{i,j-1,k}) - \alpha_3(f_{i,j,k+1} + f_{i,j,k-1}) \\ & -2(\beta_{12}f_{\xi\eta} + \beta_{23}f_{\eta\zeta} + \beta_{13}f_{\xi\zeta}), \end{aligned} \quad (21)$$

which forms a tri-diagonal system of $I - 2$ linear equations along the ξ direction.

$$\begin{pmatrix} b_2 & c_2 \\ a_3 & b_3 & c_3 \\ & O & O & O \\ & & a_i & b_i & c_i \\ & & & O & O & O \\ & & & & a_{I-2} & b_{I-2} & c_{I-2} \\ & & & & & a_{I-1} & b_{I-1} \end{pmatrix} \begin{pmatrix} f_2 \\ f_3 \\ M \\ f_i \\ M \\ f_{I-2} \\ f_{I-1} \end{pmatrix} = \begin{pmatrix} d_2 \\ d_3 \\ M \\ d_i \\ M \\ d_{I-2} \\ d_{I-1} \end{pmatrix}, \quad (22)$$

where

$$a_i = (\alpha_1)_i, \quad (23)$$

$$b_i = -2((\alpha_1)_i + (\alpha_2)_i + (\alpha_3)_i), \quad (24)$$

$$c_i = (\alpha_1)_i, \quad (25)$$

$$d_i = \begin{cases} D_i - a_i f_{i-1,j,k}, & i = 2 \\ D_i, & i = 3, 4, \dots, I - 2 \\ D_i - c_i f_{i+1,j,k}, & i = I - 1 \end{cases}, \quad (26)$$

$$\begin{aligned} D_i = & -\alpha_2(f_{i,j-1,k} + f_{i,j+1,k}) - \alpha_3(f_{i,j,k-1} + f_{i,j,k+1}) \\ & -2(\beta_{12}f_{\xi\eta} + \beta_{23}f_{\eta\zeta} + \beta_{13}f_{\xi\zeta}). \end{aligned} \quad (27)$$

Tri-diagonal eq. (22) is solved using the Thomas method (Chapra & Canale 2015) to obtain the coordinates of all the i points at the j line and the k surface. Repeating the above step from $j = 2$ to $j = J - 1$, we get the final coordinates at the k surface.

Since the shapes of the absorbing boundaries are not fixed, we apply geometric symmetry to obtain the boundary points,

$$\begin{aligned} \vec{r}_{1,j,k} &= 2\vec{r}_{2,j,k} - \vec{r}_{3,j,k}, \\ \vec{r}_{I,j,k} &= 2\vec{r}_{I-1,j,k} - \vec{r}_{I-2,j,k}, \end{aligned} \quad (28)$$

$$\begin{aligned} \vec{r}_{i,1,k} &= 2\vec{r}_{i,2,k} - \vec{r}_{i,3,k}, \\ \vec{r}_{i,J,k} &= 2\vec{r}_{i,J-1,k} - \vec{r}_{i,J-2,k}. \end{aligned} \quad (29)$$

This solution process is equivalent to solving a parabolic partial differential equation using an implicit scheme.

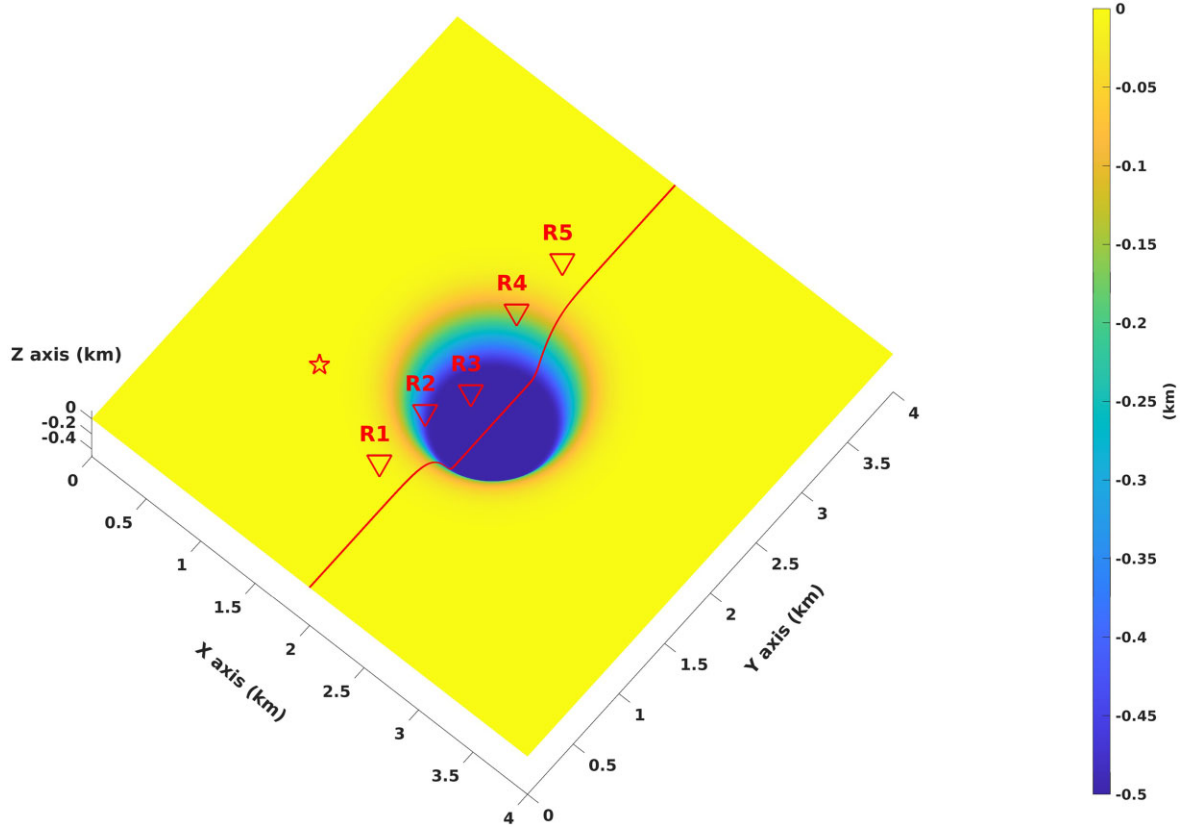


Figure 14. Illustration of the basin model topography, where the asterisk represents the seismic source, the inverted triangle represents the receivers on the free surface and the curve represents the surface position of the grid slice to be displayed.

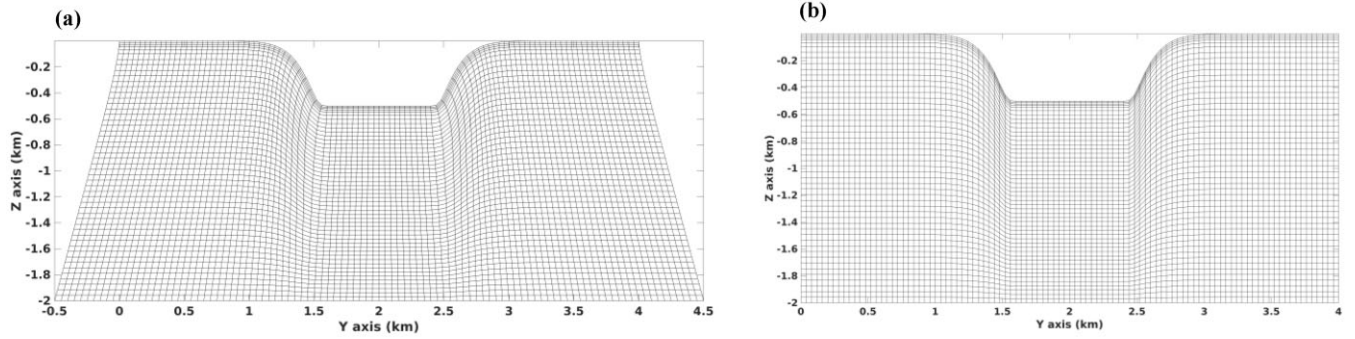


Figure 15. The basin model grid slice generation illustration. For clarity, the grid density has been reduced. (a) The curvilinear grid. (b) The vertically stretched grid.

2.4. Grid quality assessment

Grid quality assessment enables quantitative analysis of various grid quality metrics and visually highlights low-quality grid regions through visualization techniques. This paper calculates the key indicators of structured grid quality, including the Jacobian determinant, orthogonality and smoothness.

The formula for the Jacobian determinant J is:

$$J = \vec{r}_\zeta \cdot (\vec{r}_\xi \times \vec{r}_\eta), \quad (30)$$

since the grid step size in the computational domain is 1, the value of the Jacobian determinant is equivalent to the volume of the grid cell.

The formulae for orthogonality are:

$$\begin{aligned} Q_{\xi\eta} &= 90 - |\theta_{\xi\eta} - 90|, \\ \theta_{\xi\eta} &= \arccos(\vec{r}_\xi \cdot \vec{r}_\eta / (|\vec{r}_\xi| \cdot |\vec{r}_\eta|)), \end{aligned} \quad (31)$$

$$\begin{aligned} Q_{\xi\zeta} &= 90 - |\theta_{\xi\zeta} - 90|, \\ \theta_{\xi\zeta} &= \arccos(\vec{r}_\xi \cdot \vec{r}_\zeta / (|\vec{r}_\xi| \cdot |\vec{r}_\zeta|)), \end{aligned} \quad (32)$$

$$\begin{aligned} Q_{\eta\zeta} &= 90 - |\theta_{\eta\zeta} - 90|, \\ \theta_{\eta\zeta} &= \arccos(\vec{r}_\eta \cdot \vec{r}_\zeta / (|\vec{r}_\eta| \cdot |\vec{r}_\zeta|)), \end{aligned} \quad (33)$$

where the Q represents the offset between the grid angle and the 90-degree orthogonal angle, and the θ represents the grid angle. The closer the Q value is to 90, the higher the grid quality.

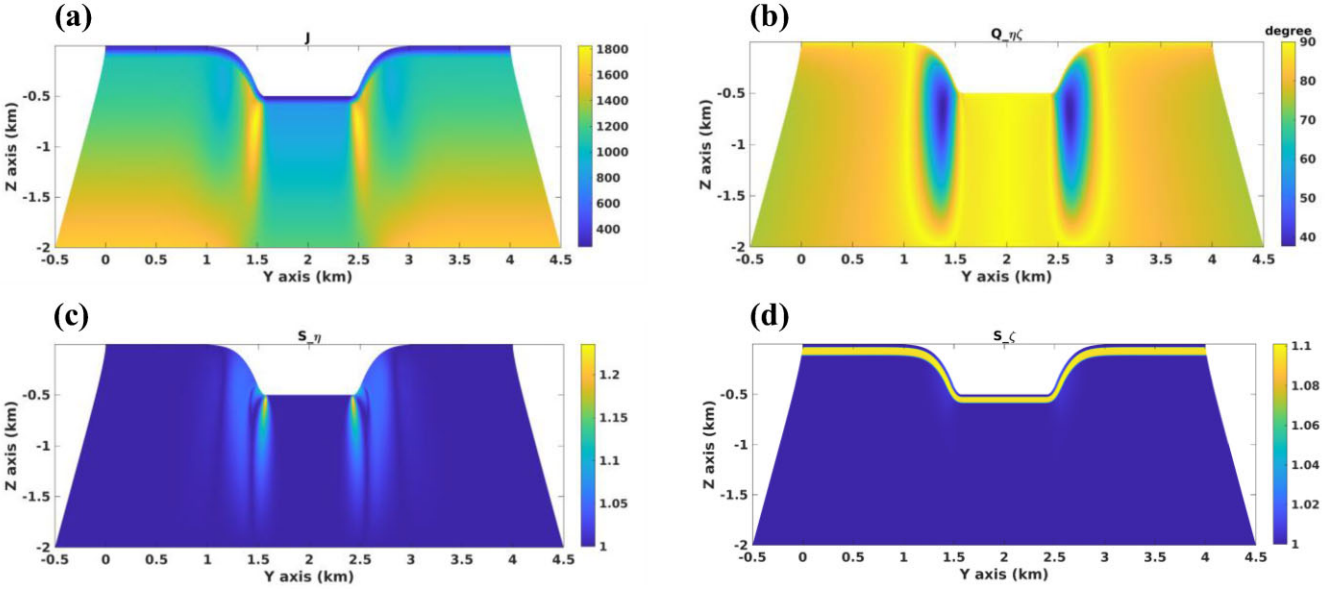


Figure 16. The basin model grid slice quality. (a) Jacobian determinant J . (b) Grid orthogonality $Q_{\eta\zeta}$. (c) Grid smoothness S_{η} . (d) Grid smoothness S_{ζ} .

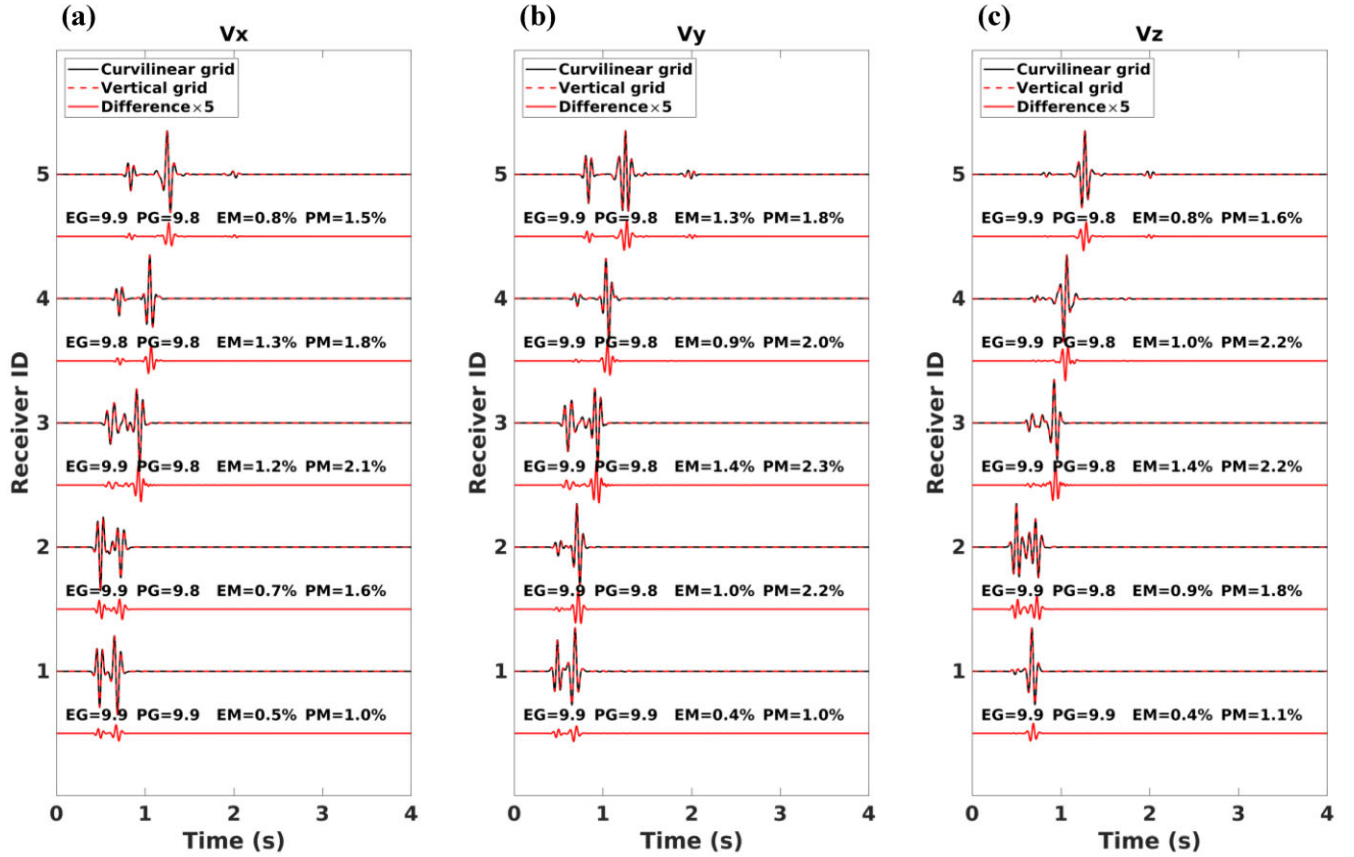


Figure 17. Comparison of the particle velocity components of the basin model at different receivers. (a) The x-component. (b) The y-component. (c) The z-component.

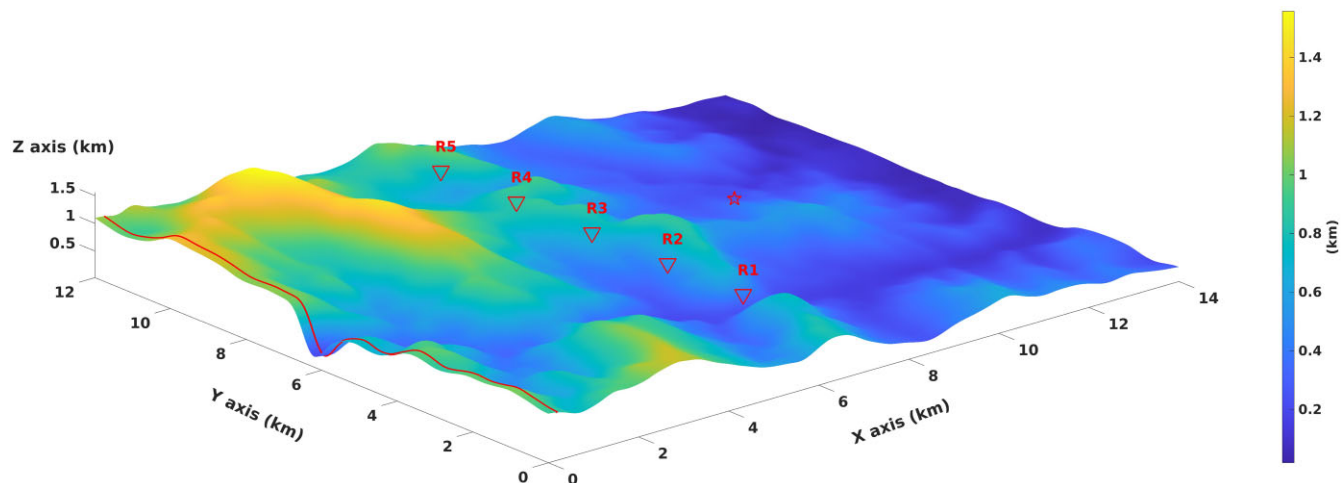


Figure 18. Illustration of the SEAM II model topography, where the asterisk represents the seismic source, the inverted triangle represents the receivers on the free surface and the curve represents the surface position of the grid slice to be displayed.

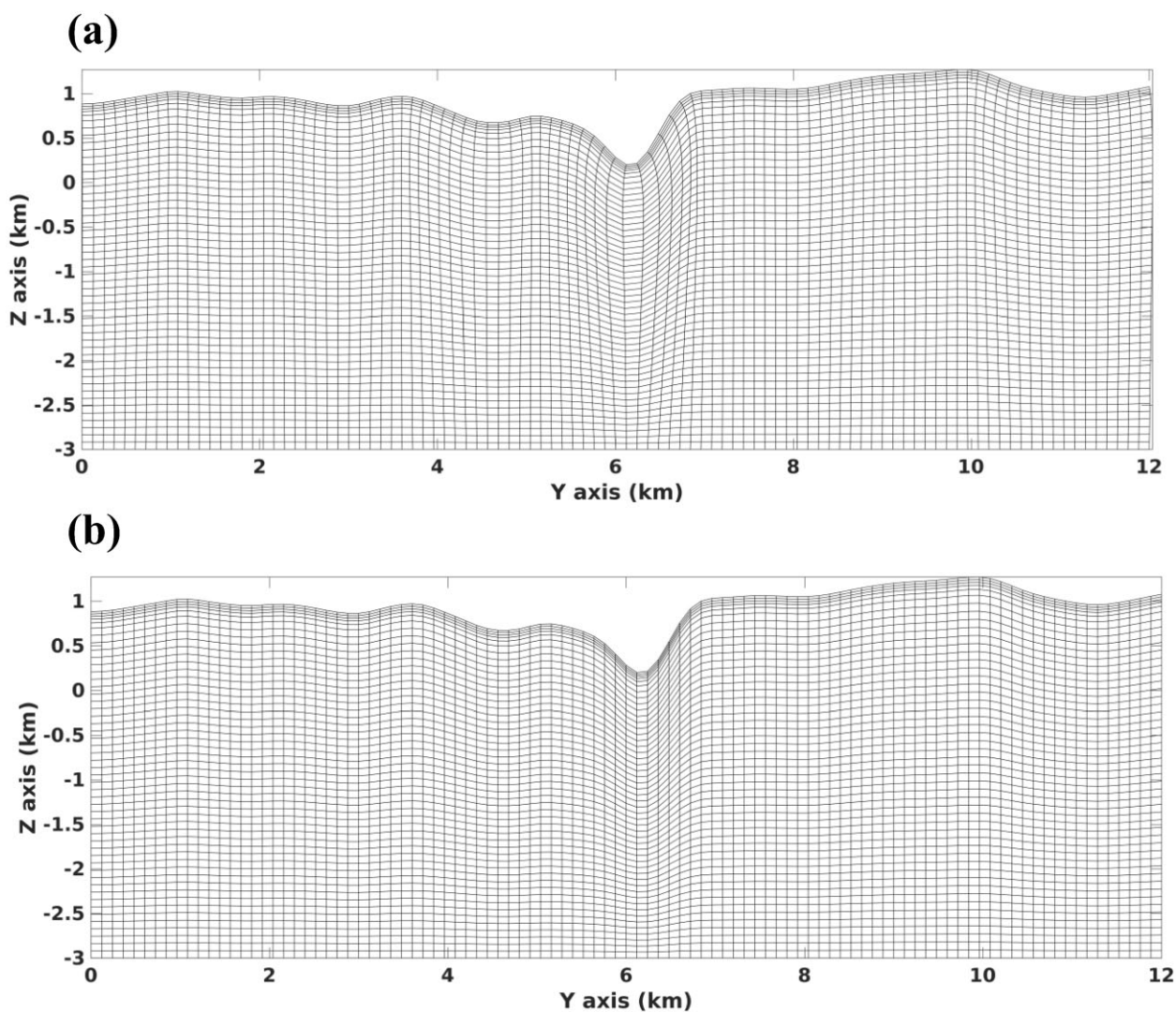


Figure 19. The SEAM II model grid slice generation illustration. For clarity, the grid density has been reduced. (a) The curvilinear grid. (b) The vertically stretched grid.

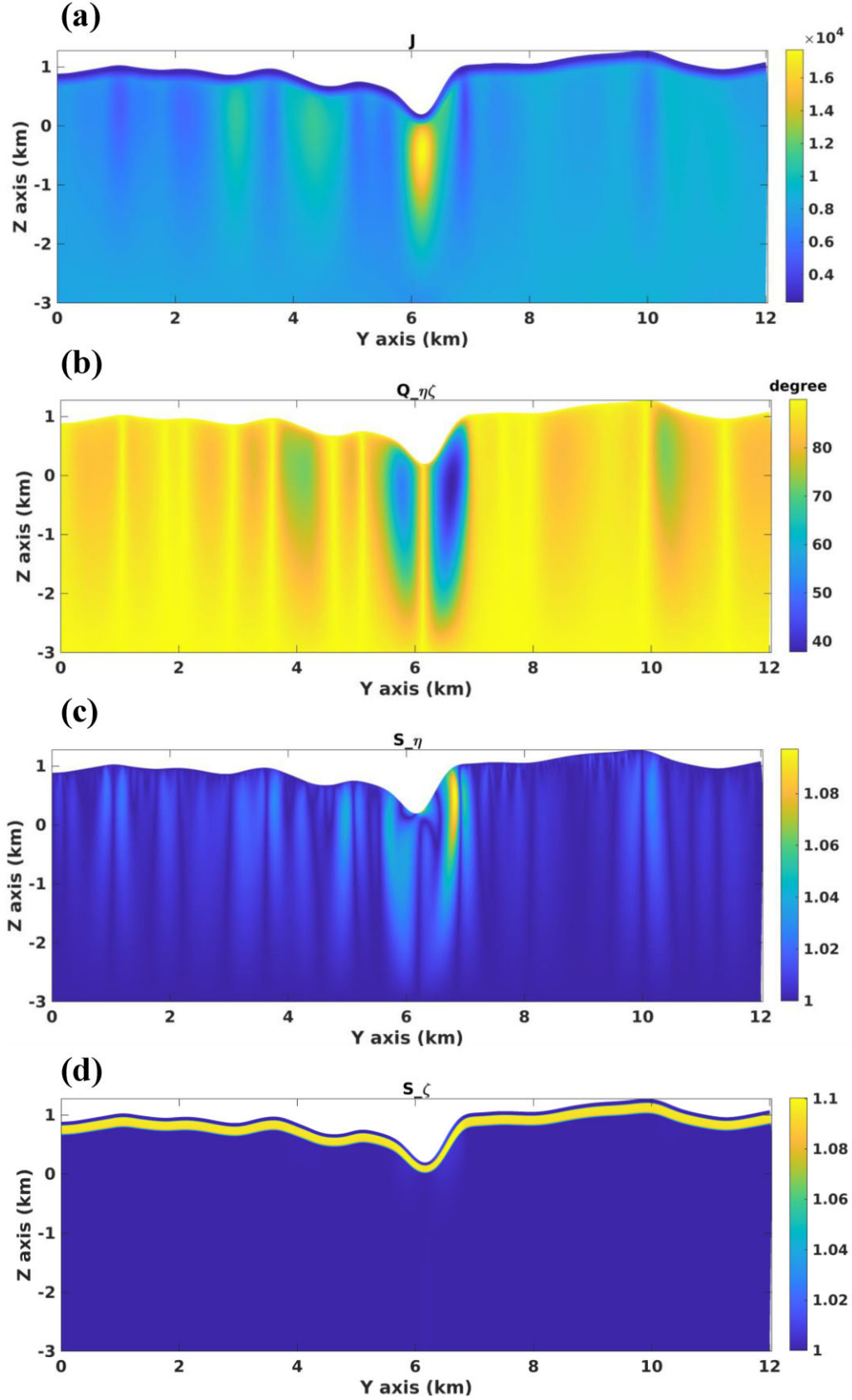


Figure 20. The SEAM II model grid slice quality. (a) Jacobian determinant J . (b) Grid orthogonality $Q_{\eta\zeta}$. (c) Grid smoothness S_{η} . (d) Grid smoothness S_{ζ} .

The formulae for smoothness are:

$$\begin{aligned} S_{\xi} &= \max(d_{\xi}^{+}/d_{\xi}^{-}, d_{\xi}^{-}/d_{\xi}^{+}), \\ d_{\xi}^{+} &= |\vec{r}_{i+1,j,k} - \vec{r}_{i,j,k}|, \\ d_{\xi}^{-} &= |\vec{r}_{i,j,k} - \vec{r}_{i-1,j,k}|, \end{aligned} \quad (34)$$

$$\begin{aligned} S_{\eta} &= \max(d_{\eta}^{+}/d_{\eta}^{-}, d_{\eta}^{-}/d_{\eta}^{+}), \\ d_{\eta}^{+} &= |\vec{r}_{i,j+1,k} - \vec{r}_{i,j,k}|, \\ d_{\eta}^{-} &= |\vec{r}_{i,j,k} - \vec{r}_{i,j-1,k}|, \end{aligned} \quad (35)$$

$$\begin{aligned} S_{\zeta} &= \max(d_{\zeta}^{+}/d_{\zeta}^{-}, d_{\zeta}^{-}/d_{\zeta}^{+}), \\ d_{\zeta}^{+} &= |\vec{r}_{i,j,k+1} - \vec{r}_{i,j,k}|, \\ d_{\zeta}^{-} &= |\vec{r}_{i,j,k} - \vec{r}_{i,j,k-1}|, \end{aligned} \quad (36)$$

where the closer the S value is to 1, the higher the grid quality. $S = 1$ means the grid is very smooth and the step size of adjacent grids is equal.

In summary, the procedure for generating grids by the parabolic grid generation method consists of the following steps:

- (1) Prepare the free surface and bottom boundary geometry file and the normalized grid step size file.
- (2) Set eq. (15) damping coefficient a .
- (3) Solving the parabolic eq. (22) to generate grids.
- (4) Measuring the quality of the grids.
- (5) Going to step 2 to reset damping coefficient a and repeat the computation until the grid quality meets the requirements.

3. NUMERICAL VALIDATIONS

To verify the effectiveness of the grid generation method, we utilized five irregular topography models for grid generation: the Gaussian peak-valley model with a uniform medium, the concave model with a uniform medium, the convex model with a uniform medium, the basin topography model with a uniform medium and the SEG Advanced Modelling Program (SEAM) II model with a complex medium (Oristaglio 2013). Seismic wave simulations were conducted based on the generated curved grid, and the results were compared with those of seismic wave simulations based on vertically stretched grids to verify the accuracy and efficiency of the generated grid in CGFDM seismic wave simulations. For all models, we implemented a 20-layer-thick perfectly matched layer (PML) (Zhang & Shen 2010) to absorb outgoing waves. Five receivers are placed on the free surface for numerical comparison of seismic wave simulations. To quantitatively assess agreement in the time domain, we calculated time-frequency misfit and goodness-of-fit measurements (Kristekova *et al.* 2006, 2009a). We report measurements of waveform agreement: single-valued envelope misfit (EM), phase misfit (PM), envelope goodness-of-fit (EG) and phase goodness-of-fit (PG). The criteria for excellent agreement are ± 0.22 for envelope misfit, ± 0.2 for phase misfit and goodness-of-fit values exceeding 8. All these values are calculated using the TF_MISFIT_GOF_CRITERIA package (Kristekova *et al.* 2009b). The seismic source is located near the free surface, exciting strong surface waves.

It should be noted that the CGFDM typically requires a PPW (points per wavelength) of at least 8 to achieve accurate body wave simulation. However, Rayleigh waves have shorter horizontal wavelengths and an exponential vertical profile. They require more PPW than body waves to reach the same level of accuracy. To address this, Zhang *et al.* (2012a) refined the grid along the depth direction near the free surface, significantly enhancing the resolution of Rayleigh waves. All the generated grids in this study were refined to meet the

Table 1. Simulation parameters for uniform medium models.

V_p (km s ⁻¹)	V_s (km s ⁻¹)	ρ (g cm ⁻³)	f_c (Hz)	t_0 (s)
3.0	2.0	1.5	10	0.2

t_0 is the time delay of the source.

requirements of the CGFDM method. We refined the top 10 grid layers below the surface. To ensure grid smoothness, we gradually transitioned from fine to coarse grids along the depth direction by a factor of 1.096. This transition zone consists of 12 grid points. The length of the coarse grid is approximately three times that of the fine grid. The seismic source is an explosive source, and the source time function is a Ricker wavelet. The central frequency f_c of the source is determined by eq. (37),

$$f_c = \frac{V_{\min}}{2.5 \times dh \times \text{PPW}}, \quad (37)$$

where $V_{\min}$ is the minimum speed, dh is the grid step. PPW (points per minimum wavelength) equal to 8, according to the requirements of the CGFDM method and the maximum frequency of the Ricker wavelet is estimated as 2.5 times the central frequency.

3.1. Gaussian peak-valley model

The grid number of the Gaussian peak-valley model is $n_x \times n_y \times n_z = 401 \times 401 \times 201$. The damping coefficient a of the generated grid is 20. The topography is shown in Fig. 2, and the grid step size of the topography surface is approximately 10 m. The bottom boundary is a plane at a depth of -2000 m, with grid points linearly and evenly distributed in the range of $x = [0 \text{ m}, 4000 \text{ m}]$ and $y = [0 \text{ m}, 4000 \text{ m}]$.

Fig. 3(a) shows the generated curved grid slice corresponding to the curve on the surface. The generated curved grid is refined along the depth direction near the free surface and is orthogonal, with the overall grid distribution being reasonable. Fig. 3(b) shows the corresponding vertically stretched grid. The grid is refined along the depth direction near the free surface, but in areas with steep terrain undulations, the grid is severely distorted and has poor orthogonality. The grid quality assessment of this slice is shown in Fig. 4. Fig. 4(a) shows the Jacobian determinant. Near the free surface, the grid is refined and remains orthogonal, resulting in small and nearly equal grid volumes in this area. The overall distribution of grid volumes is reasonable, with the maximum value being about 1600 and the minimum value being about 200. Fig. 4(b) shows the grid's orthogonality. The grid is orthogonal near the free surface, gradually transitioning to the distorted grids deeper inside. The smallest grid angle is approximately 35 degrees. Fig. 4(c) shows the grid smoothness S_{ξ} . The overall smoothness of the grid is close to 1, with the maximum S_{ξ} being around 1.2. Fig. 4(d) shows the grid smoothness S_{ζ} . The overall smoothness of the grid is close to 1, with the transition from the fine surface grid to the coarse internal grid having a value of about 1.1. The grid quality assessment results indicate that the generated grid has a reasonable volume distribution with no abnormal large or small values. Near the free surface, the grid exhibits good refinement and orthogonality, with excellent overall smoothness, making it capable of generating high-quality grids for the Gaussian peak-valley topography.

Seismic wave simulations were conducted using these two types of grids. The model and simulation parameters are provided in Table 1. The seismograms were recorded for 4 s and the time step Δt was determined by the CFL stability condition. The total time steps for the curvilinear grids are 4938, with time step $\Delta t = 8.1 \times 10^{-4}$ s.

The total time steps for the vertically stretched grids are 7407, with time step $\Delta t = 5.4 \times 10^{-4}$ s. The computation time for the vertically stretched grid is about 1.50 times longer than the curvilinear grid. Fig. 5 compares the results of the particle velocity components for both grids. The overall waveform fitting shows a high degree of consistency. To present the waveform error more clearly, we have magnified the error by a factor of ten. The goodness-of-fit values for all receivers exceed 9.9, with maximum envelope misfit (EM) of 1.0 per cent and maximum phase misfit (PM) of 0.8 per cent. For the Gaussian peak-valley model, the high-quality curvilinear grid has a significant advantage over the vertically stretched grid, as it reduces the computational time while maintaining the same seismic wave simulation accuracy.

3.2. Concave model

The grid number of the concave model is $n_x \times n_y \times n_z = 401 \times 301 \times 201$. The damping coefficient a of the generated grid is 5. The topography is shown in Fig. 6, with the grid step size on the surface approximately 10 m. The bottom boundary is a plane at a depth of -2000 m, with grid points linearly and evenly distributed in the range of $x = [-500 \text{ m}, 3800 \text{ m}]$ and $y = [0 \text{ m}, 3000 \text{ m}]$.

Fig. 7(a) shows a slice of the generated curvilinear grid corresponding to the curve on the surface. The grid is refined along the depth direction near the free surface, maintaining orthogonality, with an overall reasonable distribution. Fig. 7(b) shows the corresponding vertically stretched grid. While the grid is refined along the depth direction near the free surface, it becomes severely distorted in areas with steep terrain undulations, resulting in poor orthogonality. The grid quality assessment of this slice is shown in Fig. 8. Fig. 8(a) shows the Jacobian determinant. Near the free surface, the grid is refined and remains orthogonal, resulting in the smallest and nearly equal grid volumes in this area. The overall distribution of grid volumes is reasonable, with the maximum value being about 1100 and the minimum value being about 300. Fig. 8(b) shows the grid's orthogonality. The grid is orthogonal near the free surface and gradually transitions to the distorted grids deeper inside. The smallest grid angle is approximately 50 degrees. Fig. 8(c) shows the grid smoothness S_ξ . The overall smoothness of the grid is close to 1, with the maximum S_ξ being around 1.08. Fig. 8(d) shows the grid smoothness S_ζ . The overall smoothness of the grid is close to 1, with the transition from the fine surface grid to the coarse internal grid having a value of about 1.1. The grid quality assessment results indicate that the generated grid has a reasonable volume distribution with no abnormal large or small values. Near the free surface, the grid exhibits good refinement and orthogonality, with excellent overall smoothness, making it capable of generating high-quality grids for the concave topography.

We performed seismic wave simulations using these two types of grids. The model and simulation parameters are provided in Table 1. The seismograms were recorded for 4 s, and the time step Δt was determined by the CFL stability condition. The total time steps for the curvilinear grids are 4000, with time step $\Delta t = 1.0 \times 10^{-3}$ s. The total time steps for the vertically stretched grids are 11 765, with time step $\Delta t = 3.4 \times 10^{-4}$ s. The computation time for the vertically stretched grid is about 2.94 times longer than the curvilinear grid. Fig. 9 compares the results of the particle velocity components for both grids. The overall waveform fitting shows a high degree of consistency. To present the waveform error more

clearly, we have magnified the error by a factor of ten. The goodness-of-fit values for all receivers exceed 9.9, with maximum envelope misfit (EM) of 1.0 per cent and maximum phase misfit (PM) of 0.4 per cent. For the concave model, the high-quality curvilinear grid has a significant advantage over the vertically stretched grid, as it reduces the computational time while maintaining the same seismic wave simulation accuracy.

3.3. Convex model

The grid number of the convex model is $n_x \times n_y \times n_z = 401 \times 301 \times 201$. The damping coefficient a of the generated grid is 10. The topography is shown in Fig. 10, and the grid step size of the topography surface is approximately 10 m. The bottom boundary is a plane at a depth of -2000 m, with grid points linearly and evenly distributed in the range of $x = [0 \text{ m}, 3500 \text{ m}]$ and $y = [0 \text{ m}, 3000 \text{ m}]$.

Fig. 11(a) shows the generated curved grid slice corresponding to the curve on the surface. The curved grid is refined along the depth direction near the free surface and remains orthogonal, with an overall reasonable grid distribution. Fig. 11(b) shows the corresponding vertically stretched grid. While the grid is refined near the free surface, it becomes severely distorted in areas with steep terrain undulations, resulting in poor orthogonality. The grid quality assessment for this slice is shown in Fig. 12. Fig. 12(a) displays the Jacobian determinant. Near the free surface, the grid is refined and remains orthogonal, resulting in small and nearly equal grid volumes in this area. The overall grid volume distribution is reasonable, with the maximum value around 1300 and the minimum value around 300. Fig. 12(b) shows the grid's orthogonality. The grid is orthogonal near the free surface and gradually transitions to the distorted grids deeper inside. The smallest grid angle is approximately 30 degrees. Fig. 12(c) shows the grid smoothness S_ξ . The overall smoothness of the grid is close to 1, with the maximum S_ξ being around 1.05. Fig. 12(d) shows the grid smoothness S_ζ . The overall smoothness of the grid is close to 1, with the transition from the fine surface grid to the coarse internal grid having a value of about 1.1. The grid quality assessment results indicate that the generated grid has a reasonable volume distribution with no abnormal large or small values. Near the free surface, the grid exhibits good refinement and orthogonality, with excellent overall smoothness, making it capable of generating high-quality grids for the convex topography.

We conducted seismic wave simulations using these two types of grids. The model and simulation parameters are provided in Table 1. The seismograms were recorded for 4 s, and the time step was determined by the CFL stability condition. The total time steps for the curvilinear grids are 4000, with time step $\Delta t = 1.0 \times 10^{-3}$ s. The total time steps for the vertically stretched grids are 9756, with time step $\Delta t = 4.1 \times 10^{-4}$ s. The computation time for the vertically stretched grid is about 2.44 times longer than the curvilinear grid. Fig. 13 compares the results of the particle velocity components for both grids. The overall waveform fitting shows a high degree of consistency. To present the waveform error more clearly, we have magnified the error by a factor of ten. The goodness-of-fit values for all receivers exceed 9.9, with maximum envelope misfit (EM) of 0.6 per cent and maximum phase misfit (PM) of 0.2 per cent. For the convex model, the high-quality curvilinear grid has a significant advantage over the vertically stretched grid, as it reduces the computational time while maintaining the same seismic wave simulation accuracy.

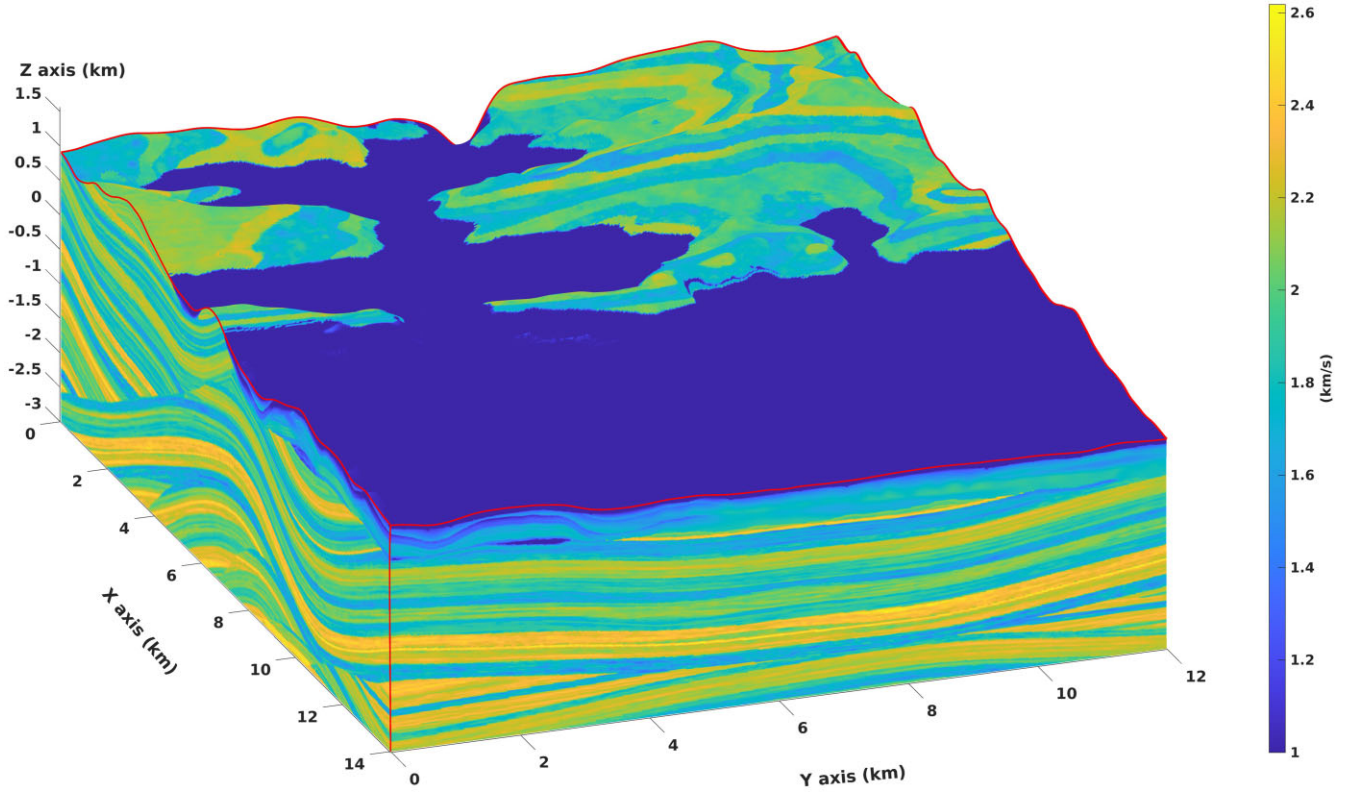


Figure 21. The SEAM II model shear wave velocity slicing of the subsurface medium.

3.4. Basin model

The grid number of the basin model is $n_x \times n_y \times n_z = 401 \times 401 \times 201$. The damping coefficient a of the generated grid is 10. The topography is shown in Fig. 14, with the grid step size on the surface approximately 10 m. The bottom boundary is a plane at a depth of -2000 m, with grid points linearly and evenly distributed in the range of $x = [-500 \text{ m}, 4500 \text{ m}]$ and $y = [-500 \text{ m}, 4500 \text{ m}]$.

Fig. 15(a) shows the generated curved grid slice corresponding to the curve on the surface. The generated curved grid is refined along the depth direction near the free surface and is orthogonal, with the overall grid distribution being reasonable. Fig. 15(b) shows the corresponding vertically stretched grid. The grid is refined along the depth direction near the free surface, but in the areas with steep terrain undulations, the grid is severely distorted, with poor orthogonality. The grid quality assessment of this slice is shown in Fig. 16. Fig. 16(a) shows the Jacobian determinant. Near the free surface, the grid is refined and remains orthogonal, resulting in the smallest and nearly equal grid volumes in this area. The overall distribution of grid volumes is reasonable, with the maximum value being about 1800 and the minimum value being about 300. Fig. 16(b) shows the grid's orthogonality. The grid is orthogonal near the free surface and gradually transitions to the distorted grids deeper inside. The smallest grid angle is approximately 40 degrees. Fig. 16(c) shows the grid smoothness S_η . The overall smoothness of the grid is close to 1, with the maximum S_η being around 1.2. Fig. 17(d) shows the grid smoothness S_ζ . The overall smoothness of the grid is close to 1, with the transition from the fine surface grid to the coarse internal grid having a value of about 1.1. The grid quality assessment results indicate that the generated grid has a reasonable volume distribution with no abnormal large or small values. Near the free surface, the grid exhibits good refinement and orthogonality, with excellent

overall smoothness, making it capable of generating high-quality grids for the basin topography.

Seismic wave simulations were conducted using these two types of grids. The model and simulation parameters are provided in Table 1. The seismograms were recorded for 4 s and the time step Δt was determined by the CFL stability condition. The total time steps for the curvilinear grids are 4348, with time step $\Delta t = 9.2 \times 10^{-4}$ s. The total time steps for the vertically stretched grids are 9524, with time step $\Delta t = 4.2 \times 10^{-4}$ s. The computation time for the vertically stretched grid is about 2.2 times longer than the curvilinear grid. Fig. 17 compares the results of the particle velocity components for both grids. The overall waveform fitting shows a high degree of consistency. To present the waveform error more clearly, we have magnified the error by a factor of five. The goodness-of-fit values for all receivers exceed 9.8, with maximum envelope misfit (EM) of 1.4 per cent and maximum phase misfit (PM) of 2.3 per cent. For the basin model, the high-quality curvilinear grid has a significant advantage over the vertically stretched grid, as it reduces the computational time while maintaining the same seismic wave simulation accuracy.

3.5. SEAM II model

The final model is the SEAM II model, which focuses on the characteristics of mountainous areas, including steep topography, soft alluvial deposits at the surface and complex subsurface structures resulting from compressive folding and thrust tectonics (Oristaglio 2013). The grid number of the SEAM II model is $n_x \times n_y \times n_z = 701 \times 601 \times 201$. The damping coefficient a of the generated grid is 10. The topography is shown in Fig. 18, with the grid step size on the surface approximately 20 m. The bottom

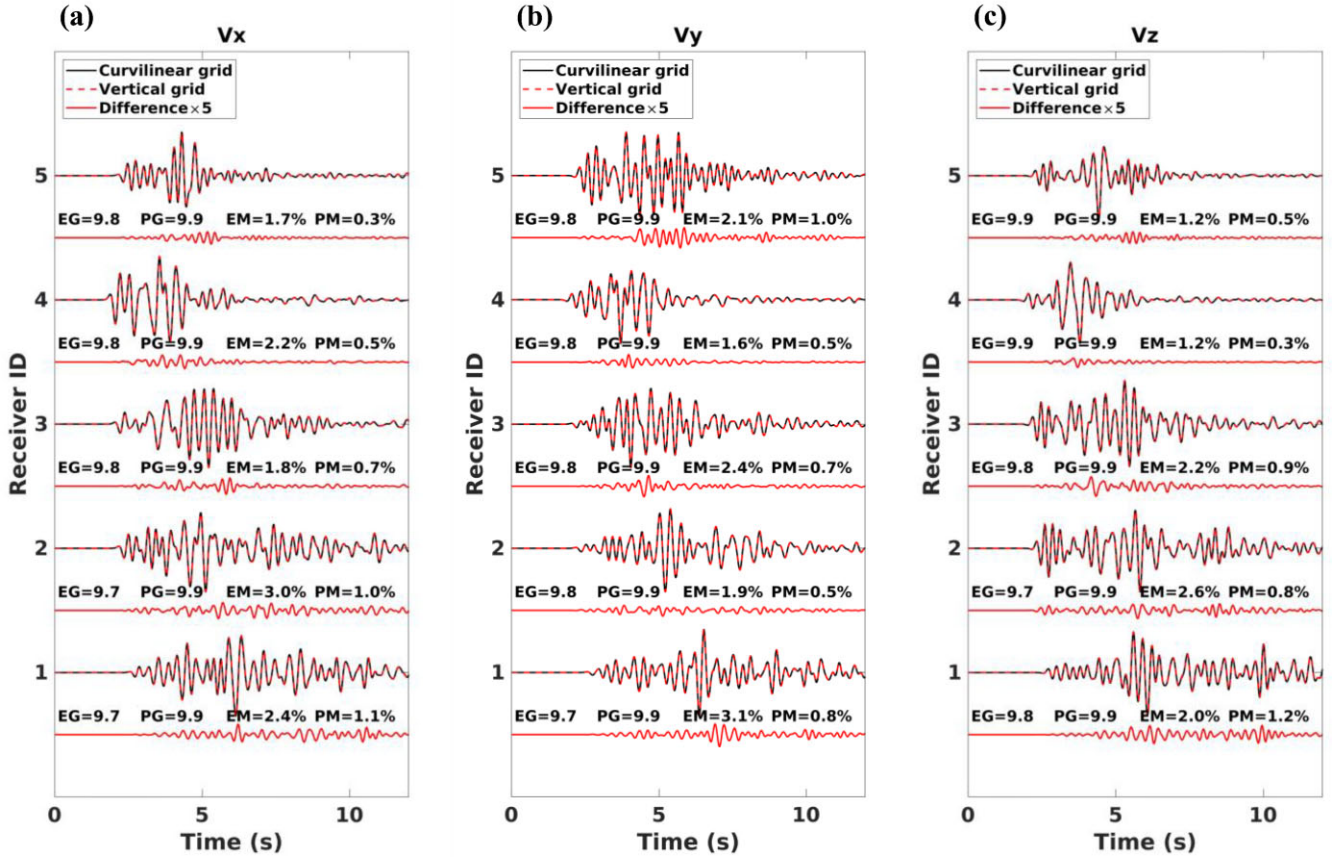


Figure 22. Comparison of the particle velocity components of the SEAM II model at different receivers. (a) The x-component. (b) The y-component. (c) The z-component.

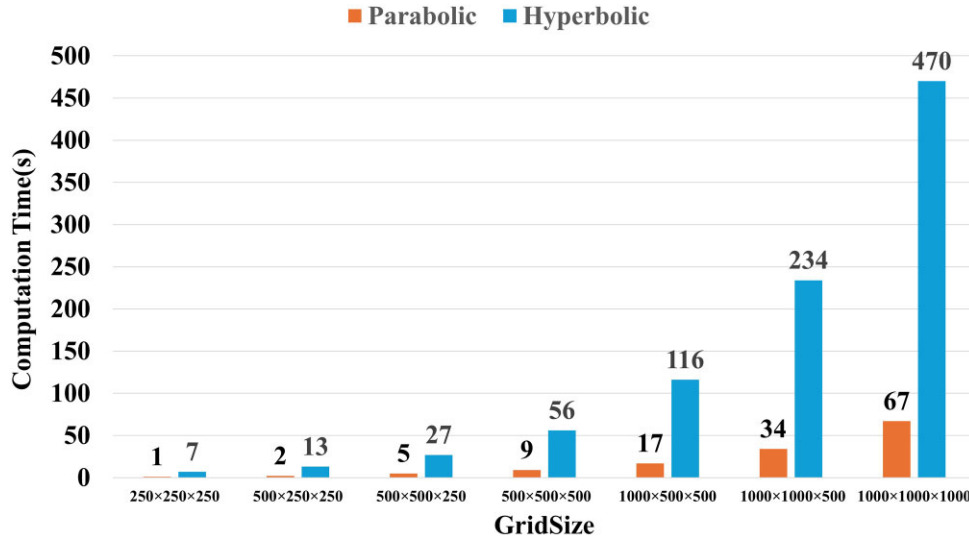


Figure 23. A comparison of grid generation times for different methods.

boundary is a plane at a depth of -3000 m, with grid points linearly and evenly distributed in the range of $x = [0 \text{ m}, 14\,000 \text{ m}]$ and $y = [0 \text{ m}, 12\,000 \text{ m}]$.

Fig. 19(a) shows the generated curved grid slice corresponding to the curve on the surface. The generated curved grid is refined along the depth direction near the free surface and is orthogonal, with the overall grid distribution being reasonable. Fig. 19(b) shows the

corresponding vertically stretched grid. The grid is refined along the depth direction near the free surface, but in the areas with steep terrain undulations, the grid is severely distorted, with poor orthogonality. The grid quality assessment of this slice is shown in Fig. 20. Fig. 20(a) shows the Jacobian determinant. Near the free surface, the grid is refined and remains orthogonal, resulting in the smallest and nearly equal grid volumes in this area. The overall distribution

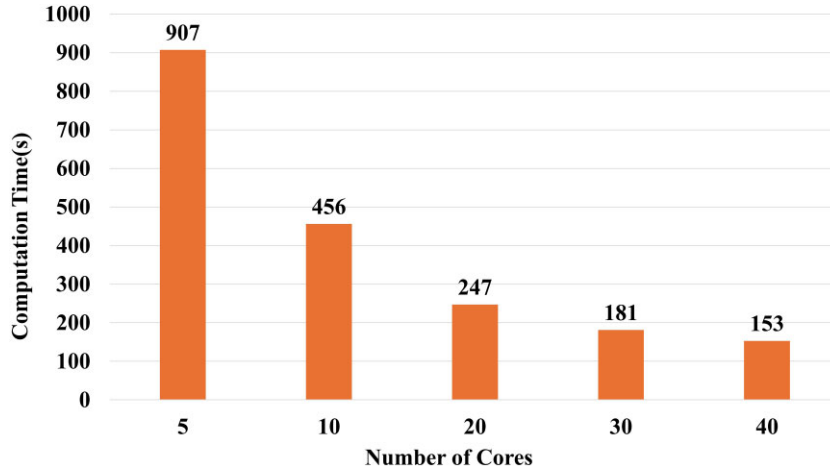


Figure 24. Grid generation time on different CPU cores.

of grid volumes is reasonable, with the maximum value being about 17 000 and the minimum value being about 3000. Fig. 20(b) shows the grid's orthogonality. The grid is orthogonal near the free surface and gradually transitions to the distorted grids deeper inside. The smallest grid angle is approximately 40 degrees. Fig. 20(c) shows the grid smoothness S_η . The overall smoothness of the grid is close to 1, with the maximum S_η being around 1.1. Fig. 20(d) shows the grid smoothness S_ζ . The overall smoothness of the grid is close to 1, with the transition from the fine surface grid to the coarse internal grid having an S_ζ value of about 1.1. The grid quality assessment results indicate that the generated grid has a reasonable volume distribution with no abnormal large or small values. Near the free surface, the grid exhibits good refinement and orthogonality, with excellent overall smoothness, making it capable of generating high-quality grids for the SEAM II topography.

Fig. 21 shows the S -wave velocity slice of the SEAM II model, displaying a complex velocity distribution. The Ricker Wavelet has a center frequency of 2.0 Hz and a time-shift of 1.0 s. The seismograms were recorded for 12 s and the time step Δt was determined by the CFL stability condition. The total time steps for the curvilinear grids are 8000, with time step $\Delta t = 1.5 \times 10^{-3}$ s. The total time steps for the vertically stretched grids are 13 043, with time step $\Delta t = 9.2 \times 10^{-4}$ s. The computation time for the vertically stretched grid is about 1.63 times longer than the curvilinear grid. Fig. 22 compares the results of the particle velocity components for both grids. The overall waveform fitting shows a high degree of consistency. To present the waveform error more clearly, we have magnified the error by a factor of five. The goodness-of-fit values for all receivers exceed 9.7, with maximum envelope misfit (EM) of 3.1 per cent and maximum phase misfit (PM) of 1.2 per cent. For the SEAM II model, the high-quality curvilinear grid has a significant advantage over the vertically stretched grid, as it reduces the computational time while maintaining the same seismic wave simulation accuracy.

4. DISCUSSIONS

To evaluate the computational efficiency of the parabolic grid generation scheme, we designed grid models of varying scales for testing. While the algebraic method generates grids very quickly, it fails to generate high-quality curvilinear grids. On the other hand, the elliptic method can generate high-quality curvilinear grids, but its computation time is extremely long, typically tens or even hundreds

of times longer than other methods. Therefore, we only compared the computation times of the parabolic and hyperbolic grid generation methods. The grid generation program is developed in the C language. The program runs on an Intel(R) Xeon(R) Gold 5220R CPU with a core frequency of 2.20 GHz. It should be noted that the computation time does not include the file input/output time. Fig. 23 presents the computation times of the parabolic and the hyperbolic methods for grid models at various scales on a single CPU. Both methods are efficient grid generation techniques based on layer-by-layer advancement, but the parabolic method is approximately seven times faster in computational efficiency than the hyperbolic method. The results indicate that the parabolic grid generation method is faster than the hyperbolic method, and the computation time increases linearly with the grid scale. For a grid with a size of $1000 \times 1000 \times 1000$, the single-core computation time is 67 s. To improve the speed and scale of 3-D grid generation, we implemented a parallelized grid generation program using the MPI method. Based on the characteristics of the algorithm, MPI performs the partitioning only in the η direction. To test the computational efficiency of running larger models on multiple cores, we designed a grid model with a size of $5000 \times 4000 \times 3000$ and performed grid generation with five different core counts, as shown in Fig. 24. As the number of cores increases, the computation time gradually decreases, but perfect linear scalability is not achieved. In the multicore scenario, our program still runs with high efficiency. For a very large grid model, when using 10 CPU cores, the generation time is 456 s, while using 40 CPU cores, the generation time is 153 s. With the support of superior hardware conditions, our program can quickly generate larger scale grids.

The curved grid is not only applicable to collocated grid finite difference methods but also to staggered grid finite difference methods. In staggered grids, physical variables are distributed at different locations within a cell, with a half-cell shifting. There are two possible ways to generate a curved staggered grid. The first method involves doubling the number of grid points; then, the odd points of the generated grid are used as the integer points of the staggered grid, while the even points serve as the half-points. The other method employs interpolation to determine the spatial coordinates of the half-grid points after generating the grid.

The parabolic grid generation method does have certain limitations and challenges in some cases. In areas with steep terrain variations (such as vertical dams or overhanging terrain), the grid may rapidly converge or even intersect. Additionally, the

distribution of grid points at the bottom boundary needs to be adjusted according to the specific topography to generate higher quality grids.

5. CONCLUSIONS

Efficient and straightforward generation of large-scale high-quality structured grids is essential for applying the FDM to seismic wave simulation in regions with irregular topography. In this paper, we developed an efficient and straightforward parabolic grid generation method to generate structured grids in such regions, based on the special feature of seismic wave simulation. This method only requires the free surface, user-specified bottom geometry and the normalized step size along the depth direction to automatically generate the grid.

The grids generated by the parabolic grid generation method start from the free surface and expand layer-by-layer along the depth direction towards the specified bottom boundary. Due to the use of a marching format for grid generation, this method is highly efficient. Moreover, it offers excellent orthogonality at the free surface. We improved the algebraic grid prediction scheme to enable flexible control of grid spacing in the depth direction. The combination of efficiency, orthogonality and flexibility makes this method an optimal choice for generating large-scale, high-quality structured grids in regions with irregular topography.

We used this grid method to generate grids for five models with irregular topography and performed seismic wave simulations using CGFDM. The five models are the Gaussian peak-valley model with a uniform medium, the concave model with a uniform medium, the convex model with a uniform medium, the basin topography model with a uniform medium and the SEAM II model with a complex medium. The generated grids feature orthogonal refinement at the free surface, with overall grid smoothness that is excellent and the volume distribution of grid elements is reasonable, indicating a high quality of the grids. Compared to conventional vertically stretched grids, the generated curvilinear grids reduce computation time while maintaining sufficient simulation accuracy. The grid generation time for large-scale models was tested, and this method can rapidly generate grids. For a model with a grid size of $1000 \times 1000 \times 1000$, the computation time on a single core is 67 s. For a large model with a grid size of $5000 \times 4000 \times 3000$, the computation time with 40 cores takes 153 s.

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DATA AVAILABILITY

The parabolic grid generation software developed in this work has been open-sourced on Github:

<https://github.com/hpc-lee/3d-grid-generate>. The waveform data used in section 3, 'NUMERICAL VALIDATIONS,' along with shell scripts that can generate the related grids using the developed parabolic grid generation software, have been archived on Zenodo: <https://zenodo.org/records/15286558>.

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